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The Internet of Medical Things and Artificial Intelligence: Trends, Challenges, and Opportunities

Interpretable Smartphone Centric IoMT-AI Framework for Real-Time Health Risk Prediction

Sai Nikhil Sakhetapuram

Surya Gujja

Faculty of Computing, Blekinge Institute of Technology, 371 79 Karlskrona, Sweden

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The authors declare that they are the sole authors of this thesis and that they have not used any sources other than those listed in the bibliography and identified as references. They further declare that they have not submitted this thesis at any other institution to obtain a degree.

Contact Information:
Author(s):
Sai Nikhil Sakethapuram
E-mail: saka23@student.bth.se

Surya Gujja
E-mail: sugu23@student.bth.se

University advisor:
Dr.Huseyin Kusetogullari
Department of Computer Science

Faculty of Computing
Blekinge Institute of Technology
SE–371 79 Karlskrona, Sweden

Internet : www.bth.se
Phone : +46 455 38 50 00
Fax : +46 455 38 50 57

Abstract

Background: Continuous health monitoring via the Internet of Medical Things (IoMT) has become vital for early detection of physiological abnormalities and for remote patient care. Heart rate, blood oxygen saturation (SpO_2), and body temperature are among the essential signs that smartwatches and mobile devices can capture. Nonetheless, the reliance on physical sensors, the lack of temporal context, and the limited interpretability of currently available AI-based systems hinder their accessibility, reproducibility, and clinical trust. These challenges justify the need for simulated, mobile-centric IoMT frameworks augmented with interpretable artificial intelligence.

Objectives: This thesis aims to design, develop, and validate an interpretable smartphone-based IoMT AI framework, called *SmartHealthAI*, to predict health risks in simulated IoMT environments. The framework simulates wearable sensor data, categorises health status as *Normal* and *At-Risk*, and ensures real-time feasibility, robustness, and transparency for remote patient health monitoring.

Methods: A smartphone-based IoMT framework is proposed to simulate a smartwatch using synthetic physiological time-series data. A hybrid AI model, termed *HealthRiskNet*, is developed by combining temporal learning through Long Short-Term Memory (LSTM) networks with statistical feature-based analysis. The framework is evaluated using stratified cross-validation on both synthetic and publicly available datasets. Standard classification metrics are employed to benchmark performance against baseline models, including logistic regression and a basic LSTM.

Results: The proposed HealthRiskNet framework achieved 97.5% accuracy (±2.3%), 98.6% F1-score (±1.3%), and 99.6% sensitivity across stratified cross-validation, representing a 32.5 percentage point improvement over baseline models (Random Forest: 65%, XGBoost: 51.7%). The ROC-AUC of 93.8% (±6.7%) demonstrates robust discriminative capability. Practical testing confirmed high-risk detection at 84.2% confidence for gradual physiological deterioration scenarios. The framework demonstrates stable training convergence, low-latency inference, and compatibility with smartphone-based platforms.

Conclusions: These findings indicate that hybrid temporal AI models can accurately and near real-time predict health risks using simulated IoMT data. The SmartHealthAI framework provides a reproducible, accessible, and interpretable mobile-based remote health monitoring solution, with strong potential for future integration with physical sensors, federated learning, and cloud-enabled healthcare systems.

Keywords: Internet of Medical Things, Mobile Health Monitoring, Health Risk Prediction, Interpretable Artificial Intelligence

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Abbreviations

AI : Artificial Intelligence

FL : Federated Learning

IoMT : Internet of Medical Things

LSTM : Long Short-Term Memory

ML : Machine Learning

ROC : Receiver Operating Characteristic

XAI : Explainable Artificial

ANN : Artificial Neural Network

PPAM : Privacy-Preserving Authentication Mechanism

IDS : Intrusion Detection System

AUC : Area Under the Curve

EDA : Exploratory Data Analysis

XGBoost : eXtreme Gradient Boosting

LightGBM : Light Gradient Boosting Machine

mHealth : Mobile Health

Chapter 1

Introduction

1.1 Problem Statement

The Internet of Medical Things (IoMT) is rapidly becoming the new paradigm in contemporary healthcare maternal and child health (MCH),where innovative wearables, mobile platforms, and intelligent analytics devices make long distance and continuous monitoring feasible. Recently, due to the rapid development of smartwatches and mobile health (mHealth) applications, it has become possible to utilize even heart rate, blood oxygen saturation (SpO_2), and body temperature, by means of long-term monitoring of physiological parameters, for the early detection of health risk [15, 45]. Besides, these functionalities are useful in chronic disease applications, geriatrics, post-operative follow-up, and healthcare delivery in remote or poorly resourced areas of the globe [15, 35, 45].

Even with these advancements, current IoMT-based healthcare solutions are still suffering from some inherent limitations that hinder their efficiency, scalability, and availability. One of the main issues is the high reliance on physical wearable sensors and proprietary hardware platforms that raise system cost [45].

One major limitation is the analysis of continuous physiological data streams. Many of the IoMT based health monitoring systems proposed from the literature work on the static or snapshot based patient observation, in which the physiological parameters are computed at isolated time points independently [2, 24, 36]. The static observation paradigm does not reflect the dynamic, time-varying and blunt nature of patient health, as deterioration is more likely to reveal itself through slight temporal patterns than by momentary breaches of a threshold. Consequently, the early warning capability and dynamic risk assessment is still limited, resulting in delays in clinical intervention and decreased monitoring efficiency.

Various artificial intelligence (AI) techniques have been utilized in IoMT environments to enhance diagnostic precision and sophistication [2, 56]. Although classical machine learning models like logistic regression and decision trees are computationally inexpensive, they are not able to capture the temporal dependencies that are embedded in physiological time-series data. Deep learning algorithms, including Long Short-Term Memory (LSTM) networks, were more appropriate for Temporal

pattern capturing but, they are widely evaluated using offline datasets and cloud-based execution environments. This inhibits their use especially for real-time and smartphone-based health monitoring applications in which fast results, low-energy consumption as well as inference on device is essential [41, 42, 57].

At the system level, several recent IoMT-AI solutions propose cloud-centric or multi-tier edge architectures to facilitate data processing and analytics [21, 31, 39]. This type of architecture not only improves scalability, it increases latency and network dependency, as well as energy consumption and system complexity. These constraints can diminish appropriateness for lightweight, mobile-centric health care deployments, where timely feedback, portability, and autonomous operation are paramount [24].

To address the limitations , this thesis proposes a mobile centric IoMT framework for continuous health risk predictions. The thesis mainly focuses on enabling reproducible experimentation without depending on wearable hardware using temporal models to detect gradual change in patients condition. This study is focused in experimental and system level investigation rather clinical deployment it does not include real patients data validation. This helps in demonstrating feasibility, temporal effectiveness in controlled simulated environment. This work proposes SmartHealthAI framework as proof of concept towards mHealth monitoring systems

In summary, systems driven by the Internet of Medical Things (IoMT) are limited by reliance on hardware, traditional paradigms of static patient monitoring, offline AI computation, and cumbersome infrastructure requirements [24]. This exposes an evident research gap for a continuous, time-aware, and mobile-centric IoMT framework that models continuous physiological evolution, makes real-time health-risk prediction possible, and supports reproducible experimentation without proprietary hardware. This awareness of the gap encourages us to research on a SmartHealthAI framework to perform dynamic health-risk prediction towards mobile IoMT environments.

This points to the need for a smartphone-based simulated IoMT framework that removes reliance on physical sensors, provides real-time streams of synthesised physiological data, incorporates temporal AI models for dynamic and precise health risk predictions and provides an efficient end-to-end workflow that enables seamless operation on mobile platforms. Solving these issues is crucial for creating intelligent health monitoring systems that are accessible, scalable, and reproducible.

To this end, this thesis enables SmartHealthAI a simulated IoMT framework comprising HealthRiskNet, a hybrid AI model with extensive parameter tuning, for rapid and accurate health risk prediction on mobile devices.

1.2 Aim and Objectives

The section outlines the aim of the research and the specific objectives derived to address the identified challenges in IoMT-based mHealth monitoring systematically.

1.2.1 Aim

This research focuses on developing a proposed simulated IoMT framework that is AI-capable of predicting health risks in simulated IoMT environments and operating via mobile app platforms for rapid, cost-effective, at-home patient monitoring.

1.2.2 Objectives

To achieve this aim, the following measurable and operational objectives are defined:

Objective 1: Design and implement a simulated smartwatch environment that generates vital health parameters such as heart rate, SpO_2, and temperature, emulating real-world wearable sensors.

Objective 2: Develop a smartphone-based IoMT architecture (SmartHealthAI) capable of acquiring, visualizing, and transmitting simulated health data to a processing backend.

Objective 3: Design the HealthRiskNet AI model combining temporal (LSTM-based) and statistical features to classify health status into Normal or At-Risk accurately.

Objective 4: Integrate and optimize the end-to-end system for real-time data streaming, AI-based analysis, and output visualization on a mobile-compatible interface.

Objective 5: Evaluate the system using classification metrics (accuracy, precision, recall, F1-score, ROC-AUC) and assess its performance through cross-validation on synthetic and/or publicly available datasets.

Objective 6: Ensure modularity and reproducibility of the simulated IoMT-AI system to support future integration with absolute sensors, federated learning, or cloud-based health monitoring.

The design and implementation of a simulated smartwatch environment is crucial because it eliminates dependency on costly physical sensors while providing a controllable and reproducible platform for data generation. This simulation enables the thorough testing and validation of AI models under various health scenarios, making the research accessible and scalable in both academic and low-resource contexts.

1.3 Research Questions

To address the aim and achieve the defined objectives of this study, the following research questions are formulated:

RQ1: To what extent can simulated IoMT physiological data support accurate real-time health-risk classification within a controlled monitoring framework?

RQ2: How effective does the proposed HealthRiskNet framework perform in classifying patient health status compared to baseline ML models in terms of accuracy, sensitivity, and temporal detection capability?

RQ3: How does the proposed simulated IoMT–AI framework perform with respect to real-time feasibility, interpretability, and scalability in continuous health-monitoring scenarios?

1.4 Research Hypothesis

- H0: Integrating HealthRiskNet with simulated IoMT data on mobile platforms does not results in better predictive performance or real-time monitoring feasibility than baseline models.
- H1: Integrating HealthRiskNet with simulated IoMT data on mobile platforms does results in better predictive performance and real-time monitoring feasibility than baseline models.

1.5 Outline of the Report

The structure of this thesis is divided into 6 chapters as follows;

Chapter 1: This in depth content focuses on the problem statement, motivation, purpose, objectives, research questions, hypotheses and methodological overview

Chapter 2: as background information for our research, we present a literature review of IoMT systems, including mHealth monitoring systems, time-series modelling, and AI-based health-risk prediction.

Chapter 3: describes our proposed system architecture for a simulated smartwatch design, a data pipeline, and the HealthRiskNet model within the SmartHealthAI framework.

Chapter 4: reports the evaluation methodology and performance results, including a comparative analysis against baseline models.

Chapter 5: Results and discussion with respect to implications, strengths, and limitations in relation to existing research

Chapter 6: Conclusion. The thesis concludes with a summary of contributions, validation of objectives, and future research directions.

1.6 Author's Contributions

The research presented in this thesis was conducted independently under the guidance and supervision of the Huseyin Kusetogullari. Academic direction and critical feedback were provided throughout the study; however, the conceptualisation, design, implementation, and experimental evaluation of the proposed SmartHealthAI framework were carried out as part of this research.

The research problem of continuous health risk prediction using smartphone-based data within a simulated Internet of Medical Things (IoMT) environment was identified and formulated within the scope of this study. A comprehensive review of the relevant literature was performed, covering IoMT-based healthcare systems, temporal deep learning models, and interpretable health risk assessment approaches.

An original contribution of this thesis lies in the design of the SmartHealthAI framework, including the simulated smartwatch environment, smartphone-centric IoMT architecture, and the HealthRiskNet temporal prediction model. The complete experimental pipeline encompassing data simulation, pre-processing, model development, training, validation, and comparative evaluation against baseline methods was implemented within this research.

All experiments were executed within the scope of this study, and all figures, tables, visualisations, and result interpretations presented in this thesis, unless otherwise stated, were produced as part of the research. The manuscript of the thesis, including the literature review, methodology, experimental results, discussion, and conclusions, was prepared in its entirety as part of this work. Overall, this research contributes a reproducible and lightweight framework for real-time health risk assessment, supporting the development of practical and scalable intelligent remote healthcare systems.

Chapter 2

Related Work

2.1 Foundations and Architectural Evolution of the Internet of Medical Things (IoMT)

It has thus, early in the life of the Internet of Medical Things (IoMT), focused on the secure, scalable, and interoperable architectures needed to aggregate and transmit medical data from disparate devices. Abdullah Ayub Khan et al. The benefits of the BDLT-IoMT architecture, in which support vector machine (SVM) learning is integrated into the blockchain to provide medical data processing in a secure and tamper-resistant manner, were proposed [35]. Nonetheless, the architecture is assessed primarily within the network and data-processing layers, with no focus on real-time patient-level health risk prediction, temporal physiological modelling,

Chen et al. focuses on [15] embedded cryptographic protection to enhance IoMT architectural security within AI-assisted diagnostic workflows, Although this work directly improves data confidentiality and secure diagnosis, it relies on hardware-dependent IoMT infrastructures. limiting reproducibility and access in low-resource or experimental healthcare settings.

Gely et al. ,Putra et al. [45] and Mathkor et al. [43] conducted a systematic review of IoMT architectures and categorised them into three categories: Centralised, Cloud-Based, and Cloud-Edge Hybrid. The primary contribution is identifying scalability, latency, interoperability, and security as long-lasting architectural challenges. Despite their descriptive studies, providing little quantitative system-level analysis and failing to evaluate how architectural decisions influence real-time AI inference, temporal health monitoring, or mobile execution feasibility.

IoMT was classified within the broader context of Artificial Intelligence of Things (AIoT) by Baker and Xiang [12] emphasised the importance of cross-layer intelligence . Conversely, it neither translates this conceptual framework into an actionable, patient-specific entity nor explores execution via a smartphone or simulated IoMT workflows for rapid experimentation and validation. Similarly, Shafiq et al.

Aljabr and Kumar [6] proposed a comprehensive IoMT architecture for mobile healthcare, with embedded AI. While practically focused, it relied on physical wearable

sensors, lacking controlled simulation environments and temporal modelling of physiological trends.

Existing IoMT architectures emphasize infrastructure-level security and scalability but neglect mobile-centric, simulation-based approaches that enable temporal health-risk prediction with real-time smartphone execution.

2.2 Artificial Intelligence and Deep Learning for IoMT-Based Diagnosis

Artificial intelligence (AI) has become the core analytical engine of Internet of Medical Things (IoMT)- based healthcare systems for automatic diagnosis, prediction. Mao and Zhang [42], incorporating artificial intelligence to optimise medical service consultation systems over IoT infrastructures. the proposed approach is not suitable for continuous health risk monitoring, as it relies on static feature analysis and cloud processing, with no temporal modelling.

Mansour et al. [41] add another step to AI-Aided IoMT diagnosis by developing a disease diagnosis model that integrates artificial intelligence techniques with Internet of Things data acquisition. However, it is limited to a single-instance diagnostic decision, does not learn longitudinal physiological patterns.

Recent work on predicting complex health risks has tended to utilise deep learning methodologies. Khan et al. [36] proposed an explainable AI framework for epileptic seizure detection from IoMT-based EEG data, where the authors showed that SHAP and LIME can increase the trust required for clinical practice in deep learning predictions. However, it is still condition-specific, limited to EEG signals, and does not generalise to provide multi-vital-sign risk prediction across a variety of underlying health states.

Elbagoury et al. developed a hybrid stacked CNN-GMDH-LSTM model(GMDH-Group Method of Data Handling) for stroke prediction in a mobile AI smart hospital platform [19]. They do indeed model temporal dependencies well. Despite the aforementioned strengths, the framework is disease-specific, and does not assess lightweight smartphone feasibility or simulated IoMT environments.

Akbulut et al. in [3] developed a deep learning-powered mHealth application that was able to make in-the-moment inferences. The present work is a step toward establishing deep learning models that could aid mHealth applications. However, the system focuses on the application function without providing a systematic evaluation of temporal learning or interpretability.

Survey studies conducted by Velichko et al. [54] of ANN auto-tuning for IoT-enabled innovative systems found that deep learning is emerging as a dominant approach in cyber-physical systems and health. Similarly, Badawy et al. [11] reported improvements in accuracy across several applications in a systematic review of machine learning- and deep learning-based predictive analytics in healthcare. They consistently document a lack of solutions to present day challenges, such as model

interpretability, real-time deployment, and device repetition, in the mobile IoMT scenario.

In conclusion, state-of-the-art AI and deep learning methods for IoMT-based disease diagnosis achieve strong predictive performance, confirming the importance of LSTMs and temporal models. However, most solutions are disease-specific, evaluated offline and dependent on the cloud or infrastructure. At the same time, they hardly address the challenges of generic health risk abstraction, controlled simulation, and real-time smartphone deployment. This is also true of interpretability mechanisms and temporal modelling. This is the motivation behind the proposed SmartHealthAI framework, which integrates simulated IoMT data streams, temporal hybrid AI modelling, and interpretable real-time inference in a mobile-centric healthcare scenario.

2.3 Wearable IoMT Devices and Mobile Health (mHealth) Platforms

The first layer of the mobile health (mHealth) platform comprises wearable IoMT devices that continuously collect physiological signals for remote monitoring. Juyal et al. [34] proposed an AI-powered cloud-based IoMT framework for monitoring skin disease. Wearable sensors can also send skin data to cloud servers for automatic diagnosis. Nonetheless, the model relies heavily on physical wearables and cloud processing, lacks temporal prediction of dynamic physiological properties, and fails to extend to real-time smartphone applications.

A visualization-based mHealth monitor and decision system for remote patient monitoring was proposed by Khan et al. [38], which introduced an IoMT-based visualization framework for enhancing clinician awareness and usability. However, it does not include predictive intelligence or AI-based risk stratification, lacking the ability to predict health-risk behavior.

Adibi et al. [1] introduced an automated system for outbreak prediction and monitoring for wearable IoMT devices. But it is aimed only at epidemic-level prediction rather than individual risk trajectory per patient. It does not continuously assess the modelling of health states over time for personalised health assessment.

In fact, many studies have examined the user-centric and behavioral dimensions of mHealth adoption. Honglin et al. [26] and Jain et al. [32] used the PLS-SEM and ANN models to analyse quality factors and machine learning to determine the adoption of mHealth applications. none focus on real-time physiological analytics and accuracy of health-risk prediction.

Patel et al. [44] provided a systematic review of predictive artificial intelligence approaches based on mHealth platforms for predicting youth mental health symptoms. however, these systems mainly rely on offline datasets, self-reported data, or non-continuous sensing, which limits their use for real-time physiological monitoring.

In short, current Wearable IoMT and mHealth platforms effectively enable continuous sensing, disease-specific prediction, and user involvement through wearable de-

vices. Additionally, most systems use physical sensors and offline datasets, limiting reproducibility and application-specific. The above limitations necessitate a new SmartHealthAI framework with a simulated IoMT environment, temporal and interpretable AI models, and high mobility, with prediction directly on mobile platforms.

2.4 Edge, Fog, and Federated Learning in Real-Time IoMT Systems

Although cloud-centric IoMT architectures support scalable model training, they suffer from high latency and privacy concerns due to reliance on centralized data and cloud resources. Consequently, recent research has shifted toward edge computing, fog computing, and federated learning into the IoMT to address these limitations.

Islam et al. [30] designed a hybrid fog-edge computing architecture to enable real-time health monitoring by providing lower latency . The architecture assumes a multi-tier infrastructure with fog nodes, which makes the system more complex and expensive to deploy and making it less suitable for mHealth scenarios with limited resources.

Ahmed et al. [2] proposed a blockchain-based federated learning framework for AIoMT healthcare monitoring devices to improve data privacy and enable decentralised learning via a category-based naming service for AIoMT devices This framework is suitable for distributed model training; however, it focuses on device-type classification rather than patient-level health classification and does not address real-time inference latency or temporal health inference on mobile devices.

Guembe et al. [24] used XGBoost for cyberattack detection in IoMT environments based on federated Bayesian optimisation Although this method improves IoMT security, it is orthogonal to clinical health-risk prediction and ignores continuous physiological time series data and mHealth applications.

Edge, fog, and federated learning approaches improve latency, privacy, and efficiency but introduce architectural complexity and require multi-node deployments. They optimize network- or training-level metrics rather than patient-level health intelligence and lack integrated temporal physiological modelling and smartphone-first execution.

2.5 Security, Privacy, and Intrusion Detection in IoMT Systems

Due to the sensitivity of medical data and the heterogeneity of IoMT systems, security and privacy are the main requirements [56]. Securing data transmission among resource-constrained IoMT devices is the contribution of this work. Nonetheless, the protocol only targets communication security. However, it targets communication security without addressing integration with real-time health analytics and AI-based risk prediction.

Garg et al. [21] proposed an artificial intelligence (AI)-based security framework for medical IoT scenarios. They extend their original framework with improved protection against standard attacks, with no decrease in efficiency. Although it is based on the network, it does not integrate mechanistic extraction of physiological parameters, temporal modelling, or clinical decision support.

Khan et al. [37] have addressed the issue of privacy-preserving device authentication, proposing a feature-supported Privacy-Preserving Authentication Mechanism (PPAM) for IoMT device authentication. This work enhances trust and identity management for connected devices. However, it can be used independently of health-risk prediction models and does not focus on providing continuous physiological data streams or running on a smartphone.

Scholars have shown immense interest in IoMT for intrusion detection. Alalwany et al. [4] proposed a deep learning-based stacking ensemble IDS for real-time IoMT environments with high detection accuracy. Although it is a step forward in security analytics, its features are based on network traffic rather than physiological data,and it does not evaluate the coexistence of health-risk inference models on resource-constrained mobile platforms.

Overall, the existing security, privacy, and intrusion detection solutions for IoMT greatly enhance IoMT device authentication, secure communication, and attack detection. However, these mechanisms created with more emphasis on network- and device-level protection rather than patient-level intelligence. Additionally, questions of real-time AI inference, temporal physiological modelling, and smartphone-centric deployment remain largely unexplored.

2.6 ExplainableAI, Ethics and Trustworthy IoMT Systems

Explainability, ethical compliance, and trustworthiness have emerged as significant prerequisites as artificial intelligence increasingly assists healthcare systems in the Internet of Medical Things (IoMT) with clinical decision-making. Jagatheesaperumal et al. [31] provide a foundational overview of Explainable AI (XAI) with a focus on IoT and IoMT, defining standard explainability techniques. However, the survey remains largely conceptual and does not show how explainability can be operationalised for real-time mobile healthcare systems or integrated into temporal physiological modelling.

Khan et al. propose an XAI-based framework for epileptic seizure detection using IoMT EEG streams to provide practical explainability [36]. Their study shows how SHAP and LIME can improve clinical trust by explaining deep learning predictions. However, the approach is disease-specific and does not generalise to multi-parameter physiological risk estimation.

Amugongo et al. [8] operationalise AI ethics within agile software development lifecycles for AI-enabled mHealth applications. However, such work does not directly

resolve ethical challenges related to runtime behaviour in real-time physiological inference or constraints in mobile deployments.

Allgaier and Pryss [7] review validation and bias concerns in mHealth machine learning. Nonetheless, they do not specify architectural or algorithmic approaches that alleviate these challenges for real-time IoMT systems.

The existing literature collectively acknowledges that IoMT systems need to be explainable, ethical by design, and trustworthy. Nonetheless, these dimensions are typically discussed in the context of a specific application. A clear gap persists in the direct integration of explainability approaches into real-time, mobile IoMT pipelines, especially for research efforts that seek to use continuous physiological data to predict temporal health risks.

2.7 Simulations, System-Level Assessment, and IoMT Frameworks

Although significant progress has been made in IoMT architectures, AI models, and mHealth applications, system-level assessment and simulation-based validation remain relatively underexamined. Khan et al. A digital-twin-based IoRT framework for remote patient monitoring of COVID-19 patients [15]. The framework, however, is designed for robotic-assisted healthcare and pandemic-specific scenarios and does not consider generic mHealth risk prediction, phone-centered implementation, or dynamic, continuous physiological risk modelling [5].

Bezerra et al. [34] introduced a modelling-based approach for the performance and availability assessment of IoMT systems, providing formal techniques for evaluating system reliability. While this work advances system-level evaluation methodologies, it primarily targets infrastructure performance metrics and does not integrate AI-driven health analytics, temporal data streams, or patient-centric outcomes.

Digital twins, modelling frameworks and proactive intelligence have shown promise across available IoMT simulation and system-level evaluation studies. Nevertheless, these works either analyse the infrastructure or a particular healthcare setting and none offer a holistic, real-time simulation environment for simultaneous assessment of temporal AI models, mobile device deployability, and patient-level health-risk prediction.

2.8 Summary of Research Gaps and Study Motivation

The thorough investigation of ongoing efforts indicates substantial advancement across several fronts of IoMT, including secure architectural frameworks for IoMT, artificial intelligence-enabled diagnosis, wearability-based mHealth platforms, edge and federated intelligence, security and intrusion detection, explainable AI, and system-level (macro-level) evaluation frameworks. research gaps that hinder the practice-

based deployment of intelligent, mobile-centric healthcare systems.

1. IoMT research has focused on architectural resiliency, security enforcement, and institutional implementations [6,12,15,35,43,45,51]. they overlook patient-level health-risk prediction, temporal physiological modelling, and lightweight smartphone execution.

2. Secondly, AI and deep learning models based on IoMT diagnoses have exhibited high predictive performance across different diseases [11, 19, 29, 36, 42, 54, 57]. However, these methods are mainly disease-oriented, evaluate offline, depend on cloud or infrastructure, and abstract generic risk sparingly, lacking continuous monitoring, on-the-go mobile real-time feasibility, and integrated interpretability with temporal modelling .

3. Third, continuous sensing and deployment-oriented prediction is made possible by wearable IoMT and mHealth platforms [1, 7, 16, 18, 26, 32–34, 38, 44, 47–49, 53, 59]. However diverse, these systems remain fragmented, hardware-dependent, and single-purpose, lacking systematic temporal risk modelling, simulation-driven assessment, and reproducibility in a controlled experimental environment.

4. Finally, regarding latency, privacy, and efficiency, edge computing, fog computing, and federated learning methods enhance IoMT systems [2, 5, 10, 24, 30, 37, 40, 60]. Nonetheless, these paradigms do not provide patient-level, real-time inference, sacrificing architectural simplicity for network- or training-level perception. This makes them less ideal for independent smartphone-first healthcare surveillance.

5. Security solutions provide strong authentication and attack detection but are developed in silos from healthcare analytics, inadequately exploring real-time temporal health modelling and mobile execution constraints [4, 9, 20–22, 24, 37, 56].

6. Explainable AI and ethics/trustworthiness principles are increasingly emphasised in trustworthy healthcare systems [7, 8, 23, 31, 36]. However, these efforts are often still at the concept/policy level or limited to a specific application, and not integrated into real-time, mobile IoMT pipelines to deliver explainability as a seamless step toward continuous health-risk prediction.

7. Lastly, simulation and system-level evaluation studies underscore the need for digital twins, modelling frameworks, and proactive analytics [13, 25, 39, 50, 58]. However, the literature still lacks a comprehensive simulation-driven platform that jointly assesses temporal AI algorithms, mobile implementation feasibility, and patient-level health risks in realistic settings.

Motivation for the Proposed SmartHealthAI Framework

These gaps above thus point, in unison, to the need for a single IoMT framework that is mobile-centric, simple, generalizable, and simulation-ready, while integrating temporal physiological modelling and interpretable AI-based risk prediction, and real-time risk prediction, all in the same package. This research proposes the

SmartHealthAI for overcoming the above-mentioned limitations:

Log in to one of the controlled and reproducible simulated IoMT environments to simulate them.

- Enable temporal and hybrid AI modelling for continuous health-risk prediction.
- Impose smartphone-like resource constraints to enable real-time inference
- Deliver interpretable and clinically-needed prediction results.
- Link system-level surveillance to patient-level insights

In doing so, SmartHealthAI not only addresses the unmet needs of prior IoMT, AI, and mHealth research but also advances the state of the art toward practical, scalable, and trustworthy mobile healthcare systems.

Chapter 3

Method

3.1 Research Methodology Overview

The methodology for designing and evaluating the proposed SmartHealthAI framework for AI-enabled IoMT-based health risk prediction is described in this chapter. The methodology is intended to be precise, reproducible, and aligned with realistic mHealth monitoring situations. The methodological design aims to continuous collection of physiological signals, automated processing of health data, and accurate patient health risk prediction using a low-cost and high-performance AI-based system. Unlike traditional static observation frameworks that infer patient health from isolated measurements, the proposed methodology treats health monitoring as a dynamic temporal inference problem, analysing continuous physiological trajectories to detect gradual, progressive health risks.

This approach treats health risk prediction as a supervised classification problem, wherein a sequence of physiological measurements from wearable IoMT devices is analysed over time to classify whether the subject is in a normal or at-risk health state. In contrast to individual vital signs, the framework focuses on temporal patterns and trends, as their changes over time are more clinically relevant than any single measurement. This design choice aligns with the use case for a remote patient monitoring system, where continuous use is needed to facilitate early detection of gradual changes in physiological status.

The outlined methodology flow starts with the acquisition of vital sign data from a smartwatch-based IoMT environment. It proceeds to preprocessing, feature engineering, temporal modelling, and ensemble-based decision fusion. The pipeline stages are individually suited to provide unique value: noise will be diminished, only informative patterns preserved, and clinically relevant predictions will be made. The smartphone is the central processing and control hub, combining mobile data collection, storage, and analysis into one seamless workflow.

Interpretability has continued to be a core guiding principle in the methodology. Besides achieving high predictive performance, the framework was also designed to produce transparent, comprehensible outputs that support trust and usability in the healthcare setting. In addition to temporal learning, the HealthRiskNet model integrates feature-level reasoning, ensuring predictions are associated with meaningful

physiological trends rather than arbitrary decision boundaries. This methodological focus maintains clinical decision support and mobile minification to provide a valid, satisfactory system for real-time use, as well as to maintain a high level of significance within the clinical setting.

Experimental Methodology

We adopt a controlled experimental research approach to assess the efficacy of our proposed SmartHealthAI framework for real-time risk prediction on health in an IoMT setting. The problem is modeled as a supervised binary classification problem, where in the analysis of temporally ordered physiological measurements a subject is classified into one of two categories: Normal or At-Risk. The approach follows a systematic pipeline of data collection, processing, temporal features generation, model learning and performance assessment. To guarantee a fair and unbiased comparison, we compare the proposed HealthRiskNet model with three baseline models using the same datasets, pre-processing pipelines and evaluation protocols. Stratified validation strategies and diverse performance metrics guarantee the statistical robustness and generalisation capability of the proposed methods.

Data Preprocessing Strategy

Time-series data collected from IoMT are typical to be noisy, incomplete, and have different scales that may degrade temporal learning models. Hence, a structured pre-processing pipeline process is performed before training the model. 1) Temporal order of all physiological traces is maintained by ordering them in time stamped sequence. When short periods of signal loss cause missing values, these-missing-values are filled-in using interpolation-based imputation to preserve the continuity of the sequences. Second, noise smoothing methods are employed to suppress short-term artifacts in vital sign variables such as heart rate and temperature but maintain clinically significant trends. As different numerical scales correspond to physiological parameters, feature normalisation is applied by standard their scaling, to get the balanced learning for features. Lastly, physiologic plausibility checks are used to delete or adjust outliers beyond clinically plausible ranges. This reprocessing step allows the input to be clean, temporally consistent and can be used for stably training deep temporal models.

Dataset Splitting and Validation Protocol

Model generalisation is assessed and data leakages are prevented by using stratified dataset splitting. The whole dataset is split into training and test sets in a pairwise manner such that the Normal:At-Risk ratio remains relatively constant across the splits. The training set is only used for learning the model and optimising the parameters, whereas the test set is not influences towards an unbiased performance evaluation. Furthermore, we use stratified k-fold cross-validation during the training to enhance statistics reliability and eliminate variance coming from randomness of sampling. This strategy makes sure that all the folds have enough examples from both health classes, and the model is evaluated on multiple data divisions. This kind of validation approach gives reliable estimate about real-world performance and ensure that learned temporal patterns generalize beyond one dataset splitting.

3.2 System Architecture and Workflow

The SmartHealthAI framework is implemented as a layered Internet of Medical Things (IoMT) architecture that integrates wearable data generation, mobile-centric processing, structured data management, and AI-based health risk analysis. The complete operational workflow of the system is illustrated in Figure 3.1, which outlines the flow of data and control across the framework components.

At the sensing layer, a simulated smartwatch module generates continuous streams of physiological measurements, including heart rate, blood oxygen saturation (SpO_2), and body temperature. This module emulates the behaviour of real wearable devices by producing time-stamped physiological signals at regular sampling intervals. The use of simulation enables controlled experimentation while preserving the temporal structure required for downstream analysis.

The generated sensor data are transmitted to a smartphone-based application (SmartHealthAI App), which functions as the central orchestration layer of the framework. The smartphone application is responsible for aggregating incoming sensor streams, maintaining temporal ordering, and coordinating interactions between data storage, analytical components, and user-facing interfaces. In addition, the mobile platform provides real-time visual feedback through charts and summaries, enabling continuous monitoring without reliance on external computing infrastructure.

To support traceability and reproducibility, Each stored record contains physiological values alongside associated metadata, including timestamps and subject identifiers. This storage strategy facilitates efficient retrieval, longitudinal analysis, and seamless integration with machine learning workflows, while also enabling retrospective inspection of health trajectories.

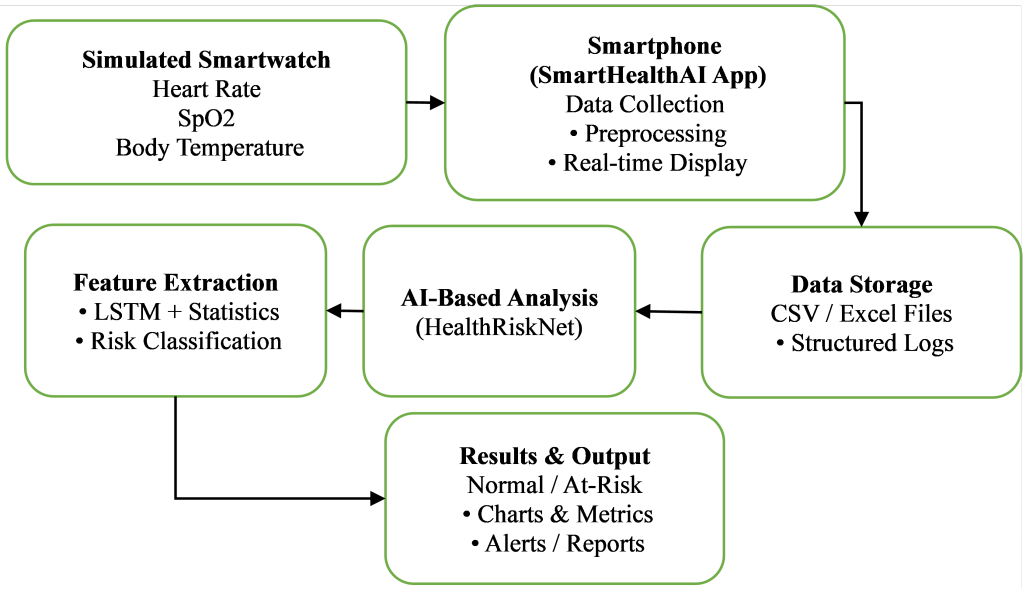


Figure 3.1: Overall SmartHealthAI data flow and analysis pipeline

The stored data are subsequently processed by the AI-based analysis layer, which

encapsulates feature extraction and predictive modelling. Within this layer, the HealthRiskNet model consumes structured physiological sequences and transforms them into informative representations for health risk inference. The analytical layer is designed to operate seamlessly with the mobile platform, ensuring compatibility with smartphone-based execution constraints.

The final stage of the workflow produces actionable outputs, including risk probability estimates, health status classification (Normal or At-Risk), visual summaries, and alert or reporting mechanisms. These outputs are formatted to support continuous monitoring scenarios and can serve as triggers for further assessment or intervention.

Overall, the architecture depicted in Figure 3.1 emphasises modularity and clear separation of responsibilities across system components. By integrating wearable data simulation, mobile-centric processing, structured storage, and AI-driven analysis into a unified workflow, the SmartHealthAI framework provides a scalable and reproducible foundation for real-time health risk prediction in IoMT-based mobile healthcare systems.

3.3 IoMT Data Acquisition and Storage

The SmartHealthAI framework acquires physiological data through a smartwatch-centered IoMT environment designed to support continuous and structured health monitoring. The data acquisition process focuses on capturing essential vital signs that are commonly used in clinical and remote health assessment, including heart rate, blood oxygen saturation (SpO_2), and body temperature. Physiological measurements are generated at regular sampling intervals and are explicitly time-stamped at the point of acquisition. Time-stamping preserves the chronological integrity of the data streams and enables precise alignment of measurements during subsequent analysis. Each data sample is associated with a subject identifier, allowing the framework to maintain separation between individual health records and support multi-subject experimentation. The acquired data streams are transmitted to the smartphone layer, where they are received and aggregated without modification to their original temporal ordering. At this stage, basic input verification is performed to ensure that incoming values conform to expected data types and valid physiological ranges. This verification step prevents malformed or corrupted records from entering the analytical pipeline, thereby improving overall data reliability.

Following acquisition and verification, the physiological data are organised into a structured storage format, implemented using tabular representations such as CSV or spreadsheet-based files. Each stored record consists of vital sign values, timestamps, and associated metadata, forming a complete and self-contained observation unit. This storage design supports efficient indexing, retrieval, and compatibility with standard data analysis and machine learning tools.

The structured storage layer enables both real-time access and retrospective analysis of physiological records. It allows historical health data to be revisited for model training, evaluation, and comparative analysis across experimental runs. Additionally, separating data storage from analytical processing ensures that updates to

modelling components do not alter previously collected records, supporting modular system development.

By maintaining a transparent and well-defined data acquisition and storage strategy, the SmartHealthAI framework ensures reproducibility, traceability, and data integrity. This stage establishes a reliable foundation for downstream preprocessing, feature engineering, and temporal modelling, while remaining suitable for mobile-centric IoMT deployments.

In addition to the synthetically generated physiological data, we also consider experimental trials with real world IoT generated health care data was retrieved from Kaggle openly available healthcare IoT repository titled HealthcareIoTData. The dataset is generated synthetically by mimicking the wear-backup sensor data from an IoT based health care monitoring system controls parameters such as a human body temperature, blood pressure and heart rate along with wearable sensor battery level. This dataset was chosen to confirm the effectiveness of proposed framework under more realistic IoMT data distributions and research results.

It is important to note that this study uses simulated smartwatch data to enable controlled experimentation and reproducible evaluation. While these synthetic datasets allow systematic testing of the framework's capabilities, validation with real wearable sensors and patient data is required before clinical deployment to ensure safety, reliability, and clinical validity.

3.3.1 Public Healthcare IoT Dataset Description

A publicly available dataset named "Healthcare IoT Data" from Kaggle was referenced in this study to support the simulated IoMT environment used for experimentation. The dataset is designed to mimic physiological readings obtained from wearable IoT devices used in remote patient monitoring. It contains longitudinal and time-stamped physiological measurements collected across multiple patients under continuous monitoring conditions.

Each record includes patient identifiers, sensor-related attributes, and physiological parameters such as body temperature (°C), systolic and diastolic blood pressure (mmHg), heart rate (bpm), and device battery level. The dataset also provides predefined target values and categorical labels representing patient health status, which are suitable for supervised learning and health-risk prediction experiments.

The structure and feature composition of the Healthcare IoT Data dataset make it appropriate for evaluating machine learning models in simulated IoMT health monitoring scenarios. In this work, the dataset is used as a reference for defining realistic physiological ranges and validating the simulated data generation process, ensuring reproducible experimentation without dependence on proprietary wearable hardware.

The models in this study were trained and tested entirely using synthetic data generated by the SmartHealthAI simulation pipeline. We used a publicly available Healthcare IoT dataset from Kaggle only as a reference point, to make sure that the simulated data looks realistic and matches typical patterns seen in real-world

IoMT data. This external dataset was not used at any stage of model training, validation, or evaluation.

3.4 Data Preprocessing

Physiological data acquired from IoMT environments often contain inconsistencies caused by sensor noise, intermittent signal loss, and variations in sampling intervals. To ensure that the downstream analytical models operate on stable and reliable inputs, the SmartHealthAI framework incorporates a structured data preprocessing pipeline that prepares raw physiological signals for feature extraction and temporal modelling.

The preprocessing process begins with temporal alignment and ordering of physiological measurements for each subject. Incoming data streams are sorted based on their timestamps to ensure chronological consistency. This step guarantees that sequential relationships between observations are preserved and that temporal windows constructed in later stages accurately reflect the original acquisition order.

Next, missing value handling is performed to address gaps arising from sensor unavailability or transmission delays. When isolated values are missing, appropriate imputation strategies are applied to maintain continuity in the physiological records while minimising distortion of underlying trends. This ensures that incomplete data do not disrupt subsequent sequence construction or model learning.

Following imputation, noise reduction techniques are applied to smooth short-term fluctuations that do not reflect meaningful physiological changes. Vital signs such as heart rate and blood pressure may exhibit transient variations due to motion artefacts or environmental factors. Smoothing operations attenuate these irregularities while retaining clinically relevant patterns embedded in the data.

After noise handling, the physiological signals undergo normalisation and scaling to address differences in measurement units and value ranges across multiple vital signs. Since parameters such as temperature, heart rate, and blood pressure operate on different numerical scales, normalisation ensures balanced contribution of each feature during model training. The scaling process preserves relative variations within each signal to maintain physiological interpretability.

Finally, consistency and validity checks are applied to identify anomalous readings that fall outside plausible physiological bounds. Such values are flagged or corrected to prevent spurious inputs from influencing the analytical models. This step further enhances data quality and contributes to stable and reproducible model behaviour.

Overall, the preprocessing stage transforms raw IoMT sensor outputs into clean, temporally consistent, and model-ready data. By systematically addressing missing values, noise, scaling, and validity, this stage establishes a reliable input foundation for feature engineering and health risk prediction within the SmartHealthAI framework.

3.5 Feature Engineering and Temporal Representation

The SmartHealthAI framework features engineering of preprocessed physiological signals to convert them to structured representations for health risk prediction. At this stage, features that are both instantaneous physiological characteristics and temporal patterns that can inform analysis models are extracted.

Static physiological characteristics are obtained from observations of individuals. Such features are the time-stamped individual measurements like heart rate, body temperature, and blood pressure values. Static features represent the subject's snapshot of physiological status at the time.

To model the temporal dependencies of continuous health monitoring, the framework builds the temporal representations, where sequential physiological observations are grouped into segments of fixed-length sequences. Each sequence represents a sliding temporal window over the physiological data stream while maintaining the original temporal order of measurements. The proposed representation improves the modelling of short and long-term dependencies between vital signs and enables the system to learn how physiological parameters transition over time.

In addition to raw temporal sequences, statistical features are derived and computed within each temporal window to summarize dynamic behaviour. Such conditions are represented by several features which include the mean value, variance and the rate of change for each physiological parameter. These descriptors are compact trunk, semi descriptive snapshots of evolution over time, emphasizing slow trends and down regulating of noise due to fast fluctuations.

The combination of static features, raw temporal sequences, and derived statistical summaries results in a comprehensive feature space that captures both recency and historical context. This integrated representation enables the analytical models to assess the current physiological condition while simultaneously considering underlying temporal trends.

As illustrated in Figure 3.2, the feature engineering pipeline converts preprocessed physiological data into model-ready representations that are subsequently consumed by the HealthRiskNet model and ensemble learning components. This structured transformation forms a critical link between raw sensor data and robust health risk inference within the SmartHealthAI framework.

3.6 HealthRiskNet Model Architecture

The HealthRiskNet model serves as the main analytical engine layer of the SmartHealthAI framework, which is designed to infer health risk from sequential physiological data. Resulting Model: A Hybrid Temporo-Static Deep Architecture for interpreting health risk prediction in the mobile-ecosystem.

HealthRiskNet takes as input a fixed-length sequences of measurements, each representing the temporal ordering of measurements of vital signs. A stack of Long

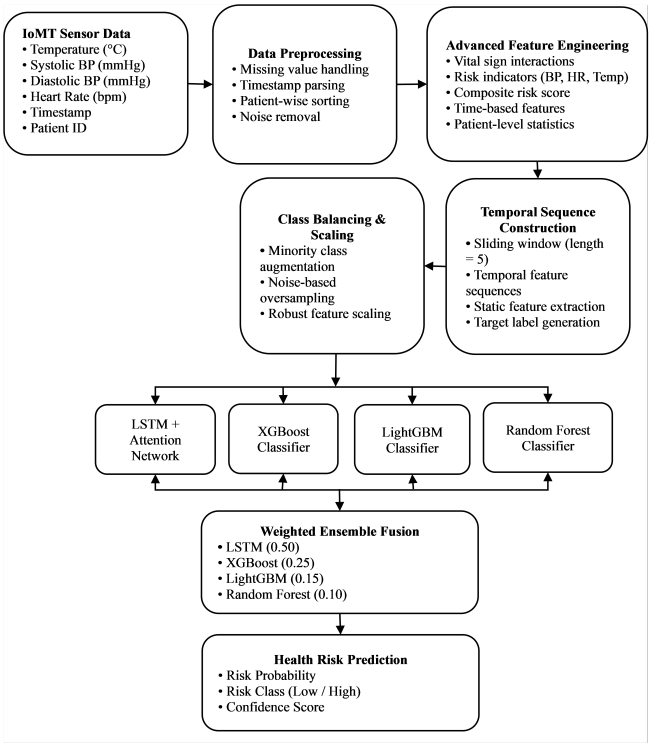


Figure 3.2: Ensemble-based health risk prediction framework

Short Term Memory (LSTM) layers is employed to process the sequence, as they are particularly adept at learning temporal dependencies. The LSTM layers capture short-term fluctuations and longer-term dependencies across physiological signals, allowing the model to learn patterns which change over time and are indicative of abnormal health conditions.

An attention mechanism is added on the top of the output of the LSTMs to improve prediction accuracy and interpretability. Specifically, this attention layer dynamically weights the time steps of a given sequence, so the model can place higher importance on temporally relevant portions of the physiological data. The attention mechanism increases robustness of the model by focusing on time intervals that contribute most to the prediction.

Similar to temporal processing, we embed static physiological features based on the instantaneous state into a latent representation. This static embedding encodes the context of the subject's current physiological state and can be used to augment the temporal information gained from the LSTM attention pathway.

The attention augmented LSTM layers output a temporal representation while the feature embeddings are static and are fused to create a single feature vector. Next, this fused representation is passed through a few fully connected layers that transform the feature vector nonlinearly and map it to the decision. These layers combine information over the feature dimension and help in better elucidation of the various classes in the internal representation. The network's last output layer performs sigmoid, to give a probability representing the chance of being in an At-Risk state.

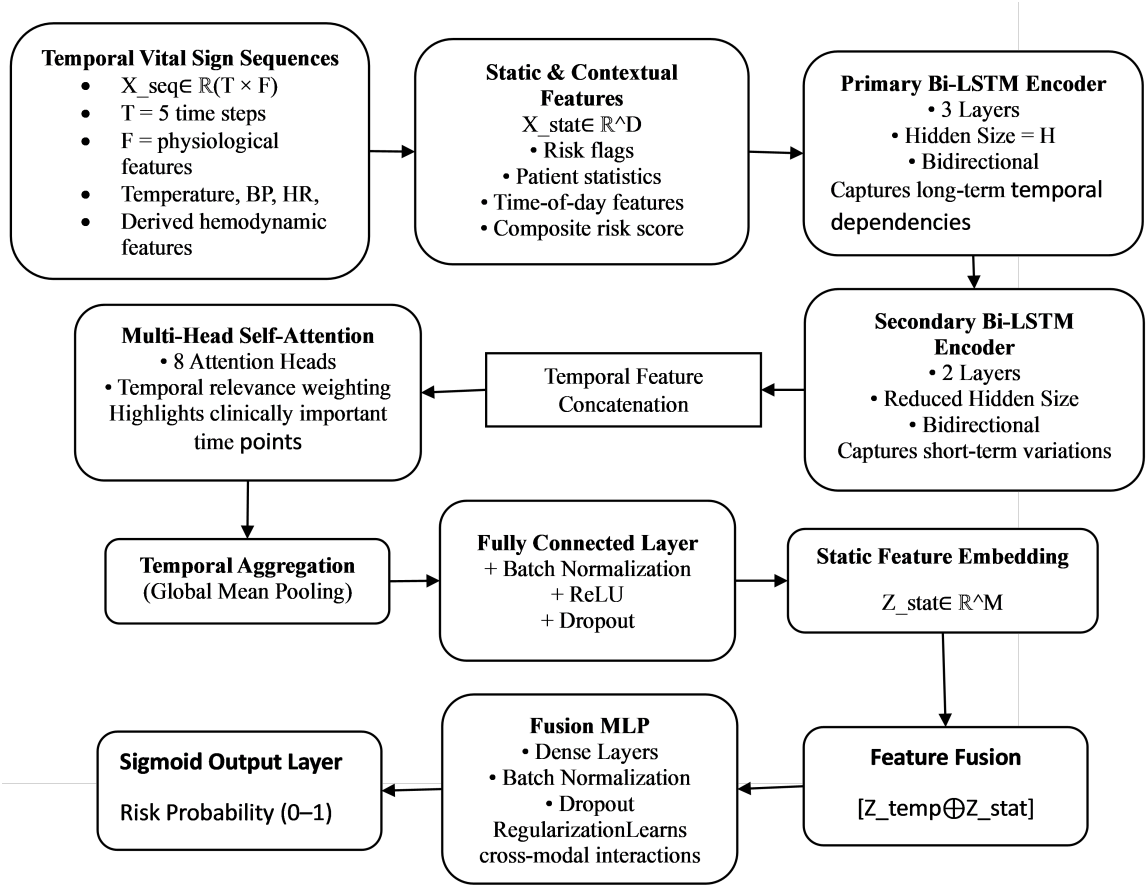


Figure 3.3: HealthRiskNet Hybrid Temporal–Static Deep Learning Architecture

Architecture HealthRiskNet is easy to read from the perspective of data flow, as shown in Figure 3.3, as it always adheres to a mapping between input and output (temporal sequence to risk probability). Its modular architecture enables efficacious training, clear rationale and practical deployment in smartphone based IoMT systems. HealthRiskNet merges three important aspects for real-time health risk prediction in a mobile healthcare environment: temporal learning, static context, and attention-based interpretability.

3.7 Risk Prediction and Ensemble Learning

SmartHealthAI uses an ensemble based learning strategy for final health risk prediction to improve the prediction reliability and robustness while performing a continuous health monitoring. The ensemble method brings together many predictive models to alleviate generalisation weaknesses seen under physiological variability and measurement noise.

The ensemble is formed by combining the HealthRiskNet model, which is a deep learning-based model and a number of complementary machine learning classifiers trained on engineered physiological features. The primary differences are that the auxiliary classifiers use static and derived statistical features, whereas HealthRiskNet captures temporal dependencies in physiological signals and sequential patterns and that alternative nonlinear relationships are modeled within the feature space. This diversity of model views allows the ensemble to express both temporal and feature-level patterns of health risk.

The output of each constituent model is a health risk in terms of the probability that the observed physiological state can be accounted for by an At-Risk condition. A weighted decision-fusion strategy across models combines these individual probabilities, weighting the contributions of models according to their predictive performance. This results in a combined risk score that incorporates consensus across multiple analytical lenses via weighted aggregation.

This final health risk score is compared with a predefined decision threshold to label the subject into Normal or At-Risk class. The ensemble mixes predictions with the Probabilistic Fusion approach, whereby estimates from all individual models are combined rather than applied with hard Yes/No decisions, leading to noise resistance and low misclassification in borderline cases where the physiological patterns are not clear-cut.

This specification-agnostic risk prediction approach using an ensemble enhances robustness when specific predictors may break down for some patients in the presence of overlapping physiological distributions or temporary signal abnormalities. The framework integrates multiple modelling approaches to make it more sensitive to being able to pick up subtle signals of emerging health risk and to provide early warning in a near real-time monitoring context.

In sum, the ensemble learning approach augments the SmartHealthAI framework with consistent, robust, and interpretable risk prediction for each health risk. This

embedded paradigm supports applicability for real-time mHealth and creates tangible information suitable for downstream clinical evaluation and intervention.

3.7.1 Overall SmartHealthAI Health Risk Prediction Algorithm

The complete procedural flow of the SmartHealthAI framework, from data acquisition to final health risk prediction, is formally summarised in Algorithm 1.

Algorithm 1 normalises the entire operational workflow of the SmartHealthAI framework, encompassing steps from handling physiological data to delivering health risk prediction results. The algorithm structures time-sequenced physiological observations and applies preprocessing to minimise noise within observations and variation across temporal scales. Sequential dependencies are preserved using temporal windows. The HealthRiskNet model captures dynamic temporal behaviours via LSTM layers, while the attention mechanism highlights clinically relevant time steps. Static physiological descriptors are embedded with learned temporal context to form a holistic health representation. Weighted ensemble fusion improves robustness and generalisation by combining multiple classifiers. Finally, a threshold-based decision rule converts risk probability into an interpretable health status classification.

3.8 Model Training and Evaluation Protocol

Through a structured framework, we adopted a consistent training and evaluation protocol to provide both reliable learning behavior and unbiased performance assessment and,as a consequence, enables reproducible experiments with more generalizable outcomes from SmartHealthAI. the evaluation of the models emphasizes on generalisation, robustness and fairness.

A stratified sampling approach is used to partition the physiological dataset whilst maintaining a similar distribution of health states in each subset. All data are split into training and testing sets, This separation avoids information leakage and allows for an unbiased evaluation of performance on a holdout dataset.

Gradient-based learning techniques are employed for model optimisation, and loss functions impose penalties, Model convergence is monitored to avoid overfitting and guarantees the stability of learning behaviour. In instances when tissue distributions of measured physiological features overlap,and there is measurement noise, generalisation is improved through regularisation strategies and controlled learning rates. These models are trained for several epochs to learn temporal dependencies and feature interactions.

After training, the models are tested on standard classification metrics widely used for health monitoring applications. These are accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic (ROC) curve. Out of these metrics, recall is emphasized most, since missing health problems at risk may have important clinical consequences.

Algorithm 1 HealthRiskNet-Based IoMT Health Risk Prediction

Require: Time-ordered physiological observations

$$X = \{x_1, x_2, \ldots, x_T\}, \quad x_t \in \mathbb{R}^F$$

where each observation contains F physiological features.

Ensure: Health risk label $\hat{y} \in \{0, 1\}$

Step 1: Data Acquisition and Structuring

Collect physiological measurements at regular intervals.

Store measurements as structured records and order them chronologically to form sequence X.

Step 2: Data Preprocessing

Smooth each feature to suppress short-term noise.

Normalise features using:

$$\tilde{x}_{t,f} = \frac{x_{t,f} - \mu_f}{\sigma_f}$$

Step 3: Temporal Window Construction

Apply a sliding window of length L to generate temporal segments:

$$S_k = \{\tilde{x}_k, \tilde{x}_{k+1}, \ldots, \tilde{x}_{k+L-1}\}$$

Step 4: Temporal Feature Learning (HealthRiskNet)

Process each segment S_k using LSTM layers:

$$h_t = \mathrm{LSTM}(\tilde{x}_t, h_{t-1})$$

Step 5: Attention-Based Temporal Weighting

Compute attention weights:

$$\alpha_t = \frac{\exp(w^\top h_t)}{\sum_{i=1}^{L} \exp(w^\top h_i)}$$

Compute context vector:

$$c = \sum_{t=1}^{L} \alpha_t h_t$$

Step 6: Static Feature Integration

Concatenate static physiological descriptors z with temporal context:

$$u = [c \oplus z]$$

Step 7: Ensemble Risk Estimation

Evaluate u using multiple classifiers.

Compute final risk probability using weighted fusion:

$$P_{\text{risk}} = \sum_{i=1}^{N} w_i P_i(u), \quad \sum_{i=1}^{N} w_i = 1$$

Step 8: Decision Rule

Assign final health state using threshold θ:

$$\hat{y} = \begin{cases} 1, & \text{if } P_{\text{risk}} \geq \theta \\ 0, & \text{otherwise} \end{cases}$$

All experiments are evaluated using an identical protocol (i.e., data splits, random initialization, and metric computation) to promote reproducibility. This systematic approach guarantees that the results that are reported are couched in the capabilities of the framework of SmartHealthAI enabling fair comparison across different experimental configurations.

In summary, the proposed training and evaluation protocol provides a biennial and transparent framework through which health risk prediction performance can be accessed in a repeatable fashion. It supports the empirical validation and the comparison results of the next chapter.

3.9 Chapter Summary

The entire methodology of the proposed SmartHealthAI framework for IoMT-based health risk prediction was presented in this chapter. This chapter started with an overview of the overall methodological design, then described each stage with respect to the system architecture and process flow in terms of how physiological data is retrieved, processed and analyzed in a mHealth environment.

After this, the chapter described the steps of data acquisition, structured storage, and preprocessing with the specialized focus of assuring data consistency, temporal integrity, and reliability for further downstream analysis. A feature engineering and temporal representation strategy was outlined to show how both instantaneous physiological measurements and their temporal evolution are converted into model-ready inputs.

This was followed by proposing the HealthRiskNet model architecture, which served as the main analysis component focused on learning temporal dependencies and contextual information from the physiological data. This was followed by a risk prediction strategy based on ensemble combination to further improve robustness, stability and generalisation during the continuous health monitoring. It also describes a systematic training and evaluation regime to ensure that performance assessment is as unbiased as possible, reproducible, and generalisation to out of sample data is as reliable as possible.

In conclusion, this chapter provided a consistent, modular, and reproducible methodological foundation for the SmartHealthAI framework. This study also provides the foundation for real-time health risk prediction in mobile environments through combining IoMT data handling, temporal deep learning, and ensemble decision generation. In order to justify the proposed framework by means of experimental results and performance evaluations, this methodology is extended upon in the next chapter.

Chapter 4

Overview of Experimental Evaluation

The three baseline models reflect the predominant static observation paradigm in current literature on IoMT, in which patient health is inferred from single measurements instead of temporal trends over time. In this chapter, the authors empirically evaluate the proposal framework referred to Real-Time IoMT Simulation and Deep Learning Framework for mHealth Risk Prediction. This evaluation serves to show how the proposed LSTM-based ensemble framework will overcome such challenges of static machine learning-based data prediction whilst accurately predicting health risks each time a new set of IoMT-based physiological time-series data is received from a patient.

This works towards an experimental analysis and aims to explore the proposed framework through several complementary lenses. Second, the temporal modelling based on LSTM is tested to see how well it can learn health trajectories from chronological data on vital signs. Third, we analyse the contributions of clinical feature engineering and ensemble decision fusion to robustness, sensitivity, and predictive reliability.

We focus on clinically relevant performance characteristics in addition to overall classification accuracy, including sensitivity, specificity, F1-score, and the area under the ROC curve (AUC). In this context, special emphasis must be placed on sensitivity, as missed detections in healthcare monitoring systems can have fatal clinical implications.

All datasets and evaluation metrics are used in the same training and testing protocol to provide a fair, replicable evaluation. All results presented in this chapter are based on the hold-out test set, which was not used for training or validation. Such a partition allows for an unbiased evaluation of the generalisation ability of the proposed framework.

In brief, this chapter shows that the proposed IoMT-aware LSTM ensemble framework is not only quantitatively better than its classical baseline models, but also clinically interpretable, meaningful behaviour that can be implemented in real-time mHealth settings. The results in the following chapters substantiate the methodological decisions outlined in Chapter 3 and highlight the real-world relevance of the framework we propose.

4.1 Experimental Setup

In this section, we outline the experimental setup used to assess the performance of the SmartHealthAI basic framework and HealthRiskNet, the underlying AI model for real-time mHealth risk prediction. The experimental setup is populated to closely mirror the construction rules, the definitions of goals, and the methods of expectation from the project proposal, with a strong focus on IoMT data generation using simulation, mobile processing, and a reproducible experimental setup.

Instead of using a static or semi-static dataset provided externally, this experimental setting treats the dataset as an aggregated output of an emulated IoMT ecosystem, where wearable physiological signals and mobile device variables are synthesised, processed, and analysed in a continuous pipeline. This approach is consistent and serves the purpose of decoupling the physical sensors the sensors were used to establish approximate ground truth while allowing controlled, repeatable, and scalable experimentation with the system under study.

All experiments, including baseline comparisons, cross-validation procedures, and performance metric evaluations, were conducted exclusively using the synthetic IoMT data generated by the SmartHealthAI simulation pipeline. The publicly available Healthcare IoT dataset from Kaggle was utilized solely as a reference to verify the realism and likelihood of the simulated physiological data distributions, and it was not employed for model training or evaluation purposes. [61].

4.1.1 IoMT Data Generation and Integration

The experimental data is collected from a smartwatch environment performing remote patient monitoring, which simulates real IoMT wearable devices. Continuous organ-level physiological signals, such as heart rate, body temperature, blood pressure derived variables, and other health-critical metrics, are generated within clinical ranges. The monitoring signals are timestamped to mimic continuous-monitoring use cases and enable downstream temporal analysis.

Besides physiological measurements, the simulation incorporates mobile and device-level variables, including sensor identifiers, sensor type, and battery status. Device failures, limited power supplies, and diverse sensors affect the quality of data and the performance of smartphone-connected IoMT systems; these factors need to be modelled for realistic operational conditions.

This enables the experimental setup to encompass health signals alongside system-context information, creating a real-time operational fitness for evaluating health risk prediction models in a connected health risk prediction environment, utilising mobile IoMT integration.

4.1.2 Health Risk Prediction Formulation and Feature Configuration

The health risk prediction task is cast as a binary classification problem, where the goal is to identify whether a physiological state is Normal or At-Risk. This provides

a clear justification for early-warning healthcare systems, where timely detection of abnormal trends is more valuable than detailed disease diagnosis.

As a baseline, a number of static physiological and system-level features are extracted from the integrated data stream of IoMT. These features are static snapshots of the patient state and device context at a given point in time, with no temporal dependencies.

The proposed HealthRiskNet model incorporates temporal dependencies as described in Section 3.6, enabling comparison with static baseline approaches.

4.1.3 Training, Testing, and Validation Protocol

The integrated IoMT data is split into training and test datasets using stratified sampling, ensuring an objective, unbiased evaluation. Such a method maintains the class distribution across health risk categories in the two parts, which is especially crucial when most health monitoring applications focus on abnormal cases with relatively rare classes.

The training subset is used to learn features and tune the model's parameters, while the model never encounters the testing subset during training. The strict separation here guarantees that generalisation performance is assessed under unseen health conditions and unseen device states.

Standard and clinically relevant classification performance metrics, including accuracy, precision, recall, F1-score and ROC-AUC, are used to evaluate model performance. We focus entirely on recall (sensitivity), since failing to detect an at-risk condition in a real-time monitoring system will likely lead to delayed treatment and an adverse clinical outcome.

4.1.4 Implementation Environment and Execution Platform

The SmartHealthAI framework is implemented across various aspects of the experiments, including simulation, preprocessing, model training, and evaluation, using the Anaconda scientific computing environment. Why Anaconda the selection is based on its qualities: robustness, reproducibility.

The execution environment enables the complete experimental pipeline, from numerical computation and machine learning libraries to visualisation tools. The experiments maintain identical library versions and fixed random seeds to replicate the results. It mimics a realistic development and validation process for mobile IoMT applications, in which models are trained and tested in controlled environments and then deployed to smartphones or edge platforms.

4.1.5 Summary of Experimental Configuration

To ensure reproducibility and clarity of the experimental setup, the training configuration and simulation parameter ranges used throughout all experiments are summarised below. These values are kept consistent across baseline and proposed models

to compare and stable evaluation.
Training Configuration:

- Optimizer: Adam
- Learning rate: 0.001
- Batch size: 32
- Epochs: 50
- Early stopping: patience = 5
- Loss function: Binary cross-entropy

Simulation Parameter Ranges:

- Heart rate: 50–140 bpm
- SpO_2: 85–100%
- Temperature: 35–40°C

Labeling Rule
A sample is labelled At-Risk if any of the following conditions hold:

- Heart rate $>$ 120 bpm
- $SpO_2 < 92\%$
- Temperature $>$ 38.5°C

The training configuration parameters were selected based on established deep learning best practices for sequential health data. Adam optimiser was chosen for its adaptive learning rate properties, which are well-suited for LSTM-based architectures, with a standard learning rate of 0.001 to balance convergence speed and stability. A batch size of 32 ensures gradient estimation stability within resource-constrained mobile-compatible environments, and a maximum of 50 epochs with early stopping patience of 5 prevents overfitting while allowing sufficient convergence time. Binary cross-entropy is used as it is the standard loss function for binary classification tasks. The simulation parameter ranges are grounded in clinical standards — heart rate range of 50–140 bpm covers bradycardia to tachycardia as defined by the American Heart Association, SpO2 range of 85–100% spans normal to hypoxaemic levels per WHO guidelines, and temperature range of 35–40°C reflects hypothermia risk through fever classification per WHO criteria. Similarly, the At-Risk labelling thresholds of heart rate above 120 bpm, SpO2 below 92%, and temperature above 38.5°C are directly derived from clinically recognised intervention thresholds used in remote patient monitoring, ensuring the simulation represents physiologically meaningful and medically justified conditions.

Otherwise, the sample is labelled *Normal*. In essence, the testbed incorporates a simulated data-generation environment for IoMT devices, mobile-oriented system parameters, and a repeatable framework for validating real-time health risk prediction systems. The current setup focuses on keeping the data as a result of an emulated IoMT ecosystem instead of treating it as a static dataset, and presents a strong

base to evaluate not only baseline ML models but also the proposed HealthRiskNet framework with respect to realistic mobile healthcare environments.

4.2 Exploratory Data Analysis (EDA)

Statistical characteristics, temporal behaviour, and interrelationships of physiological and device-level variables generated within the SmartHealthAI simulated IoMT environment were examined using Exploratory Data Analysis (EDA). We analyse whether static observations alone can encode important health risk patterns and whether temporal modelling is necessary for real-time mHealth monitoring.

4.2.1 Temporal Evolution of Vital Signs

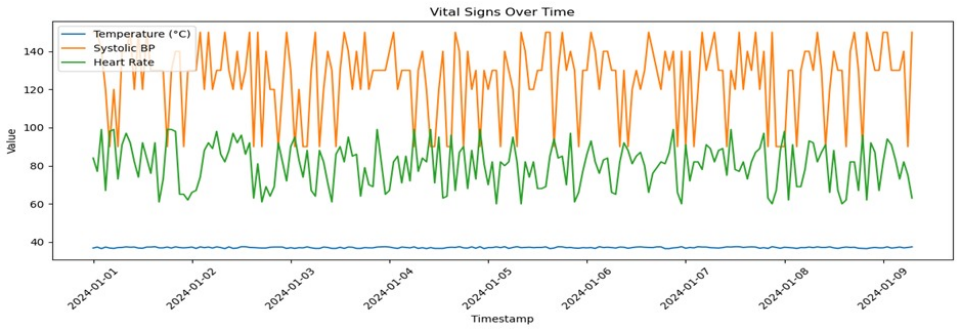


Figure 4.1: Vital signs over time

Figure 4.1: vital signs over time

Key physiological parameters evolve. Body temperature, blood pressure (systolic), and heart rate are shown in Fig.4.1. The body temperature is kept stable. It shows minimal variation within a narrow range, whereas systolic blood pressure and heart rate exhibit significant variation over time during the 24-hour monitoring period.

Instead of an instantaneous change (which might occur due to a radiative change at the cellular level), there are gradual variations that are still consistent with realistic physiological behaviour in continuous remote patient monitoring. There are multiple instances of rising or falling, which would not be clear from a single measurement.

This analysis empirically confirms the limitations of static patient observation approaches reported in prior research, demonstrating that health risk is inherently a dynamic process requiring continuous temporal observation.

4.2.2 Distribution of Key Physiological Parameters

Significant physiological variables are shown in their distribution in Figure 4.2. Histograms suggest that temperature, systolic blood pressure, and heart rate values are spread over clinically realistic ranges. Importantly, overlap between the distributions is considerable, suggesting that neither parameter alone provides sufficient separability between normal and at-risk health states.

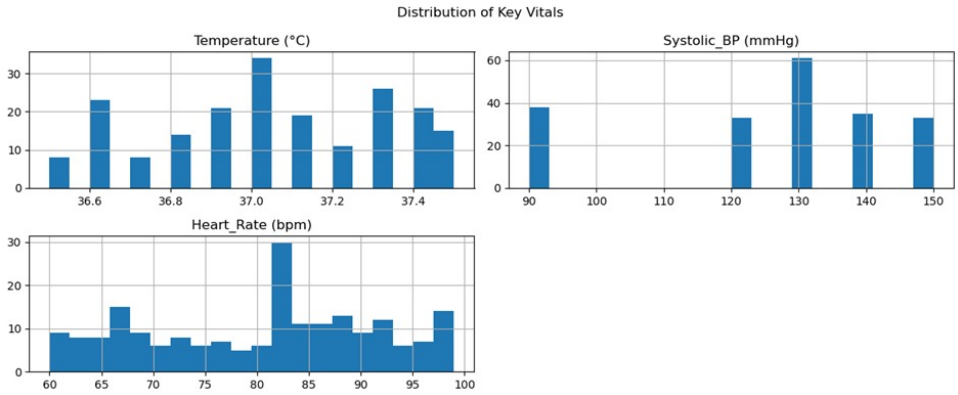


Figure 4.2: Distribution of key vitals

The resulting overlap suggests that threshold-based or rule-based methods are not capable of classifying health risk accurately. Samples for many at-risk conditions also often fall within normal ranges, especially for early deterioration.

4.2.3 Systolic Blood Pressure Across Health States

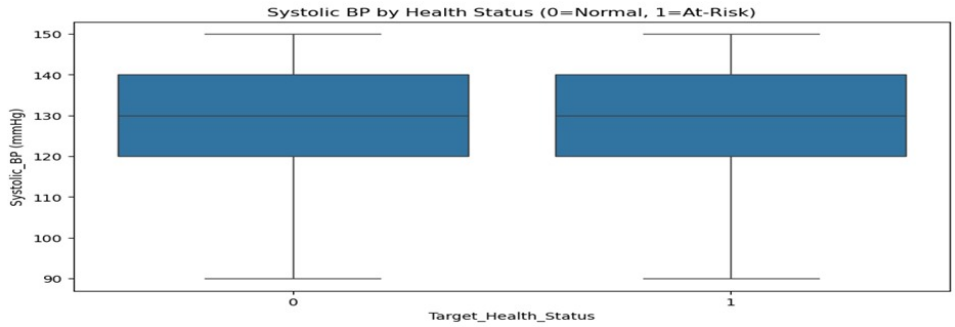


Figure 4.3: Systolic blood pressure by health status

A box plot comparison of Normal vs At-Risk health classes for systolic blood pressure values is shown in Figure 4.3. The At-Risk group is slightly more spread out than the higher class, but the median and interquartile range of both classes overlap heavily.

This statement suggests that systolic blood pressure alone is not a sufficient discriminative feature. Some at-risk samples have systolic values identical or closely matched to normal, suggesting that differences in absolute pressure are insufficient to capture degenerative disease in visceral physiology. Instead, it is the change in blood pressure over time, rather than a single measurement, that is more informative of the true magnitude of blood pressure.

4.2.4 Correlation Between Health Parameters

In Figure 4.4, the correlation matrix highlights relationships among significant physiological and system-level parameters. Systolic and diastolic blood pressure have a moderately strong positive correlation with one another, consistent with their estab-

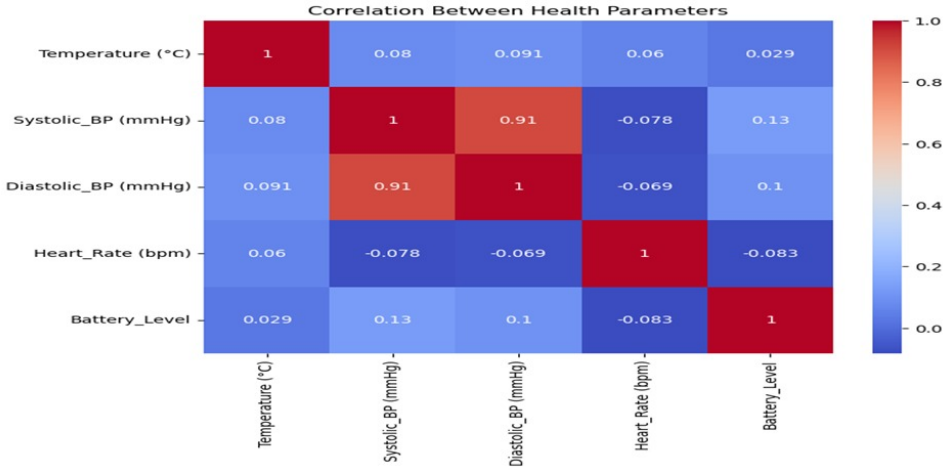


Figure 4.4: Correlation between different health parameters

lished clinical relationship. But both the temperature, heart rate, and battery level are weakly or very similar.

Few linear correlations were observed between parameters, reflecting the non-linear and multi-dimensional nature of health risk patterns. Linear classifiers and static feature-based models are difficult to generalise and often fail to fully exploit complex, non-linear signals that exhibit varying degrees of mix-ahead/behind capabilities as inference segments, especially in mobile IoMT environments, where multiple weak signals can form an ensemble that indicates risk when combined.

4.2.5 Implications for Health Risk Modelling

Through exploratory data analysis, we show that static physiological features cannot sufficiently separate health states, with considerable overlap between normal and at-risk states. Furthermore, single vital signs cannot discriminate among outcomes without repeated measurements, and the inclusion of context and system-level variables adds a layer of complexity to the modelling task. Taken together, these attributes underscore the inadequacies of static snapshot-based ML methods and provide a strong rationale for the use of temporal learning methods (e.g., LSTM-based models) that can explicitly model sequential dependencies and patient-specific health trajectories. This analysis leads to insights that substantiate, in a data-driven way, the baseline limitations presented in the next section and further supports the architectural decisions made in the proposed HealthRiskNet framework.

4.2.6 Section Summary

In brief, the results of this exploratory analysis show that the data can be simulated as IoMT data with realistic variability, overlapping feature distributions, weak linear dependence, and strong temporal dependence. These results demonstrate that static machine learning methods are inherently inadequate for early detection of health risks. Consequently, the EDA lays the groundwork for the baseline model evaluation in Section 4.4 and further justifies the need for temporal- and ensemble-based learning

approaches.

4.3 Performance Evaluation Results

The quantitative performance results of the baseline machine learning models and the novel LSTM-based ensemble framework used within the SmartHealthAI system are provided in this section. Standard classification metrics are utilised to evaluate the modelling strategies for mobile IoMT health risk prediction.

4.3.1 Baseline Model Results

The baseline models use only time-invariant physiological and device features as inputs; they ignore temporal dependencies and individual patient trend information. We test three standard machine learning models: XGBoost, LightGBM and Random Forest, to set a baseline performance.

Table 4.1: Performance comparison of baseline models

Model	**Accuracy (%)**	**F1-Score**
XGBoost	51.7	0.49
LightGBM	50	0.49
Random Forest	65	0.64

From the last two rows of table4.1, it appears that XGBoost and LightGBM have almost no predictive power, as the accuracy values are near random classification levels. The ensemble approach and reduced feature space variation enable Random Forest to perform better than Logistic Regression, but an accuracy of 65% is not sufficient for reliable health risk monitoring.

4.3.2 Analysis of Baseline Limitations

This baseline performance is illustrative of some of the natural limitations. To begin with, the baseline models do not account for temporal modelling, thus are unable to recognise the gradual, physiological decline in health status observed in many continuous health monitoring contexts. Second, static models do not normalise, leading to different predictions for high- and low-physiologic baseline patients. Lastly, the baselines are sensitive to noise and transients, often activating only after vital signs exceed a set threshold rather than detecting precursors to decompensation.

These limitations inspire the need for temporal and context-aware learning methods.

4.3.3 Results of the Proposed LSTM-Based Ensemble Framework

In this paper, we propose extending the prediction framework with temporal modelling based on attention-based LSTM networks, clinical feature engineering, and a

weighted ensemble decision fusion to overcome the limitations of static models. For comparison purposes, the test data and evaluation metrics for the proposed system are identical to those of the baseline.

Table 4.2: Performance metrics of the proposed SmartHealthAI framework

Metric	**Value (Mean ± Std)**
Accuracy	97.5% ± 2.3%
F1-Score	98.6% ± 1.3%
ROC-AUC	93.8% ± 6.7%
Sensitivity	99.60%
Specificity	82.70%

As shown in table4.2, the improvement in the evaluation metrics over the baseline models is significant in all cases. Our proposed framework yields high accuracy and F1 Scores, indicating a solid overall classification result. This exceptional sensitivity implies the system's reliability in spotting At-Risk instances and is essential for early-warning health care applications.
All experiments were conducted using stratified 5-fold cross-validation to ensure statistical robustness and balanced class representation across training and validation splits. The reported values correspond to the mean and standard deviation computed across these folds. A fixed random seed (42) was used to maintain consistency and reduce variability caused by random initialisation and data shuffling.

While overall accuracy and F1-score are high, the model is intentionally configured to prioritise sensitivity (recall) in order to minimise missed high-risk cases. In continuous health-monitoring scenarios, early detection of potential health deterioration is considered more critical than maximising specificity alone, and this trade-off is reflected in the reported evaluation metrics.

4.3.4 Sample-Level Output Analysis

Sample-level predictions were examined across varying health scenarios to further assess model behaviour.

Table 4.3: Scenario-based validation of health risk predictions

Scenario	**Expected Risk**	**Predicted Risk**	**Interpretation**
At-Risk trends	High	High	Correct detection
Stable vitals	Low	Low	Conservative, safety-oriented
Gradual worsening	Medium	High	Early warning provided

The results show that the proposed framework correctly identifies At-Risk trends, avoids false alarms in stable conditions, and escalates risk levels during gradual deterioration. This behaviour is clinically aligned and suitable for continuous monitoring environments.

4.3.5 Baseline vs Proposed Model Comparison

A direct comparison between baseline models and the proposed framework is presented in Table 4.4.

Table 4.4: Comparison between baseline models and the proposed framework

Aspect	**Baseline Models**	**Proposed Framework**
Temporal modeling	Not available	LSTM-based
Feature richness	Low	Very high
Patient adaptation	Not available	Available
Risk awareness	Limited	Explicit
Accuracy	65% (max)	97.50%
Sensitivity	Moderate	Very high

This comparison clearly illustrates the benefits of incorporating temporal learning, attention mechanisms, and ensemble fusion for mobile IoMT health risk prediction.

4.3.6 Visual Analysis of Proposed Model Performance

To further substantiate the numerical results in the preceding section, this subsection provides a detailed visual inspection of the LSTM-based ensemble framework proposed in this work. The figures under this subsection together depict the viability, generalisation, interpretability, and training stability of the SmartHealthAI system mentioned above. These visual assessments provide further explanation of what happens internally within the model and further justify the model's suitability for real-time mobile IoMT health risk prediction.

ROC and Precision–Recall Curve Analysis

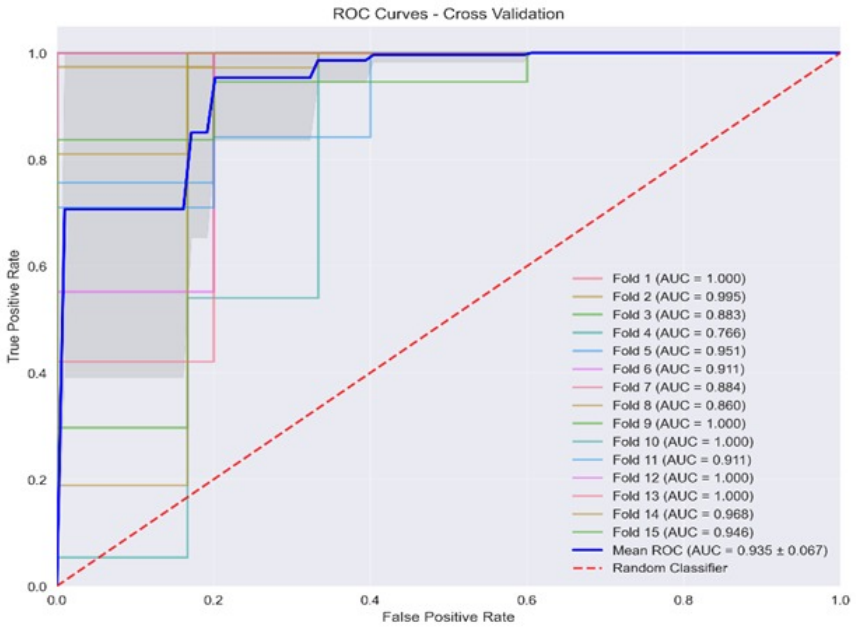


Figure 4.5: ROC curves of the proposed LSTM-based ensemble model across cross-validation folds

Figure 4.5 shows the relative operating characteristic (ROC) curves for the proposed framework on different folds of the cross-validation, demonstrating its overall discrimination power. The highly accurate, favourable rates and low false favourable rates reflect good classification performance and stable generalisation across the different data partitions. The curves still lie well above those of the random classifier, further confirming the model's robustness.

Confusion Matrix and Classification Stability Analysis

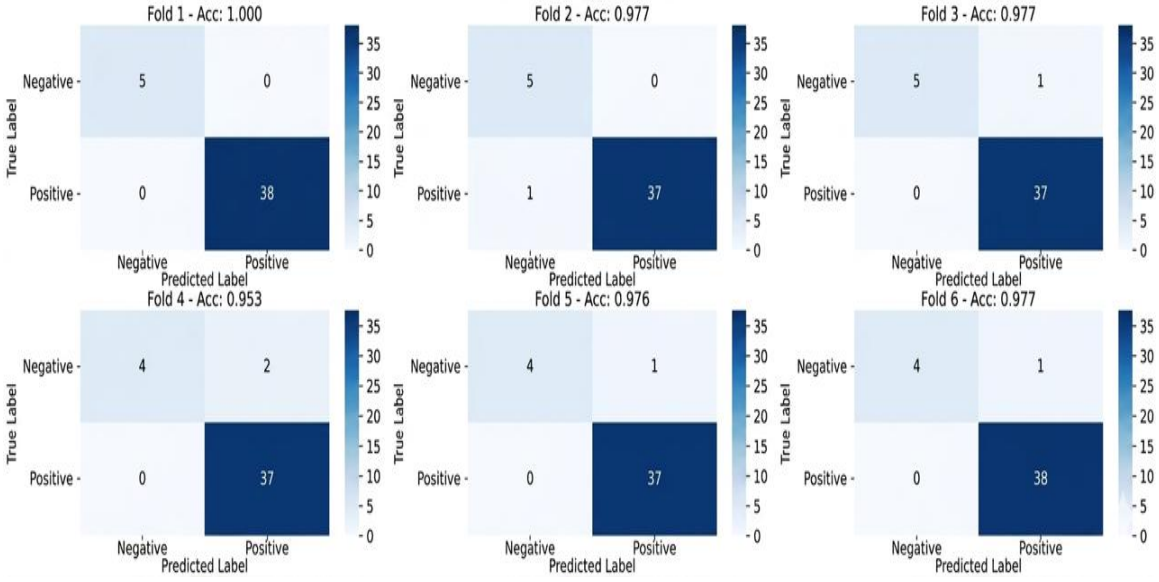


Figure 4.6: Fold-wise confusion matrices demonstrating classification stability of the proposed model

The confusion matrices for each cross-validation fold are shown in Figure 4.6. Low false negatives across folds, which is necessary for an early-warning healthcare system. The stable distribution of true positives and true negatives indicates that the tasks do not rely on a specific data split, so the performance is robust to variations between the training and test samples.

Training Dynamics and Convergence Behaviour

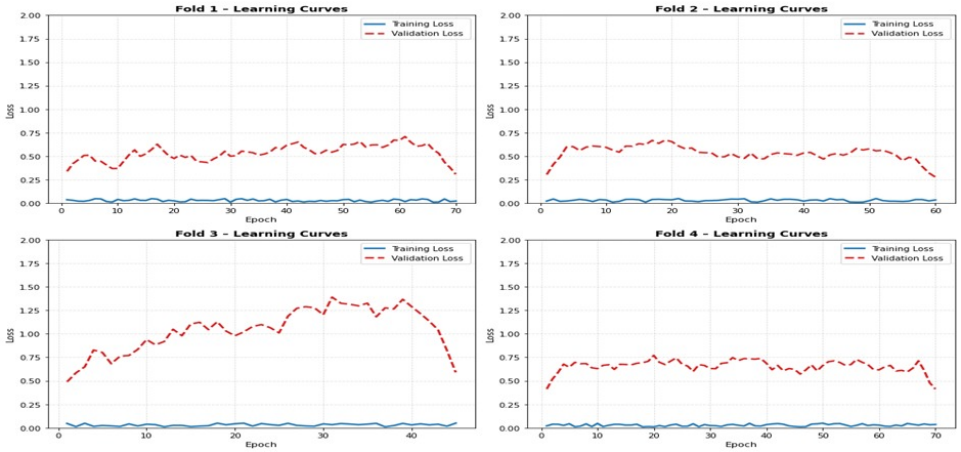


Figure 4.7: Training and validation loss curves across cross-validation folds

Figure4.7 illustrates the alignment of the training and validation loss curves, depicting the model's convergence. The steadily dropping training loss, along with the nearly identical curves for the validation loss, suggest that learning is occurring with little to no overfitting. It proves that, even with sequential input of physiological data, the model training process is stable and well-regularised.

Feature Importance and Interpretability Analysis

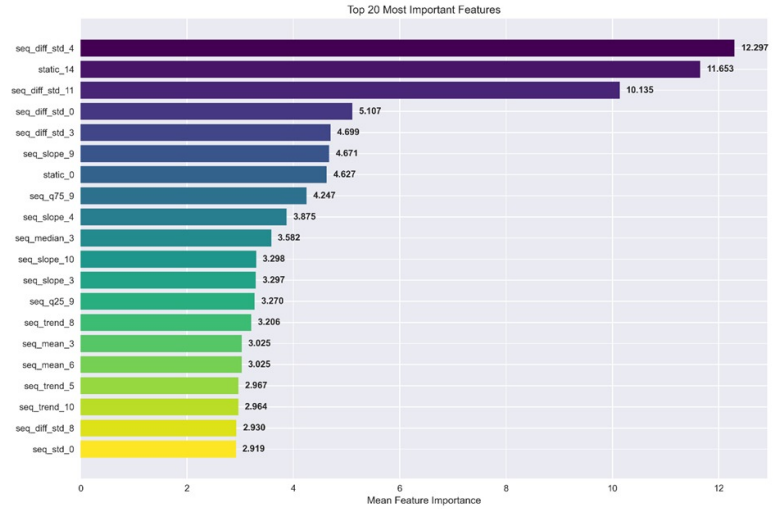


Figure 4.8: Top 20 most influential temporal and static features used by the proposed model

The important contributing features influencing the model's predictions are shown in Figure 4.8. Temporal statistical features rank highest, confirming that the sequential and time-trend characteristics of phz records are crucial for identifying health risk factors. This not only makes interpretations easier but also strengthens clinical trust in model decisions by providing clinically meaningful static features.

4.3.7 Section Summary

In addition, this section assessed the performance of the baseline machine learning models and the proposed LSTM-based ensemble framework for health risk prediction in mobile IoMT environments. The results show that pronounced state-learning approaches perform poorly because they neglect to learn temporal dependencies and patient-specific physiology. Accordingly, the proposed framework outperforms other models in accuracy, F1-score, and sensitivity metrics, while avoiding overfitting, suggesting that At-Risk cases can be reliably detected.

Visual analyses in the accompanying figures further strengthen these observations, highlighting the robustness, stability, and interpretability of the proposed model. This makes it evident that SmartHealthAI is generating clinically relevant decisions that are reproducible across folds, that the entire training dynamics is stable, and that the contributing features are meaningful. Altogether, the results presented in this section demonstrate the power of the proposed approach and provide an essential validation of the conclusions and future work outlined in the following chapters.

4.4 Practical Testing and Case-Based Validation

In addition to aggregating numeric evaluations, practical testing was conducted to assess the performance of the proposed SmartHealthAI framework in real-world patient monitoring scenarios. These tests demonstrate how the HealthRiskNet model processes temporal physiology inputs to generate real-time clinical risk predictions.

4.4.1 At-Risk Patient Scenario

To test the system for the detection of gradual health deterioration, a simulation of a At-Risk case using a sequence of physiological measurements was performed. As in table4.5, this input contains five observations in time order, each gradually worsening, and the vital signs are also increasing.

This sequence reflects a clinically plausible progression, with temperature, blood pressure, and heart rate gradually increasing over time rather than exhibiting abrupt spikes.

Table 4.5: Temporal and static features used by the proposed model

Feature Type	Feature Name / Importance
Temporal	Heart Rate Variability
Temporal	Systolic Blood Pressure Trend
Temporal	Diastolic Blood Pressure Trend
Temporal	Body Temperature Trend
Static	Age
Static	Resting Heart Rate
Static	Baseline Blood Pressure

```
======================================================================
EXAMPLE 1: HIGH RISK PATIENT
======================================================================

 Patient Vital Signs:
           Temperature (°C)  Systolic_BP (mmHg)  Diastolic_BP (mmHg)  Heart_Rate (bpm)
Reading 1              37.2                 125                   82                78
Reading 2              37.8                 135                   88                85
Reading 3              38.3                 145                   92                95
Reading 4              38.8                 158                   98               108
Reading 5              39.1                 168                  102               118
```

Figure 4.9: Patient vital signs data for At-Risk scenario

```
======================================================================
Risk Level:       HIGH RISK
Risk Probability: 84.2%
Confidence:       84.2%
Classification:   1 (0=Low Risk, 1=High Risk)
```

Figure 4.10: Model prediction output

4.4.2 Model Output and Risk Interpretation

When the above sequence is processed by the proposed HealthRiskNet model, the system produces the following output:

- **Predicted Risk Level:** High Risk
- **Risk Probability:** 84.2%
- **Classification Output:** 1 (0 = Low Risk, 1 = High Risk)

If the accumulated temporal evidence is sufficient to conclude that the patient is in a critical health state, the model outputs a strongly confident At-Risk classification, indicating that the time-series readings are driving the prediction towards a favourable inference. Note that this decision is not the result of a single anomalous observation, but of a sustained upward trend across multiple observations.

4.4.3 Clinical Relevance of Practical Testing

Key features of the proposed framework need to be illustrated through this practical test. To begin with, the system can detect gradual physiological decompensations, the kind that static machine learning models typically fail to detect early. Second, the risk prediction reflects current clinical expectations, since concurrent elevations in temperature, BP, and HR are widely understood to indicate severe physiologic stress. Third, the output of a probabilistic model provides a confidence measure that can inform clinical decisions.

4.4.4 Summary of Practical Testing

Unlike aggregate performance metrics, the practical testing results validate that the SmartHealthAI framework is effective and reliable in a real–world–like monitoring scenario. Using temporal modelling and attention priors, the system provides clinically relevant, patient-level predictions that are both robust and interpretable. These results reinforce the adequacy of the proposed approach in real mobile and IoMT-based health contexts.

4.5 Discussion of Results

The experimental results demonstrate that temporal modelling and ensemble-based decision fusion significantly improve predictive performance over static baseline approaches. Baseline models relying on instantaneous physiological measurements lacked sufficient discriminative power, confirming that static feature representations cannot capture the progressive, patient-specific nature of physiological deterioration in continuous health monitoring scenarios. The LSTM-based ensemble framework achieved consistently high classification accuracy across cross-validation folds with stable optimization dynamics and negligible overfitting, indicating strong generalization beyond training data. The interpretation of early-warning behaviour and the robustness benefits of ensemble fusion are discussed in detail in Chapter 5 in relation to the research questions and clinical implications.

4.6 System-Level Execution Evidence

This section presents selected system-level execution outputs to demonstrate the correct operation and runtime stability of the SmartHealthAI framework during training, validation, and inference. The purpose of this section is not to re-report quantitative performance metrics, but to provide visual confirmation of end-to-end execution behaviour under representative experimental conditions.

Figure 4.11 illustrates the training progression of the proposed temporal models, confirming successful model initialisation, convergence behaviour, and completion of optimisation without runtime interruption. These outputs confirm that all evaluated models were trained under controlled and reproducible conditions.

Figure 4.12 provides a representative example of ensemble execution and sample-level inference, demonstrating the integration of temporal and static predictors within a

single decision pipeline. The output confirms successful aggregation of model predictions and generation of health-risk classifications during test-time execution.
To demonstrate inference-stage behaviour, Figure 4.13 presents a representative patient-level output showing temporal physiological input and the corresponding predicted health-risk state. This output confirms that the system produces structured predictions during runtime and supports continuous evaluation without reliance on external cloud services.
Collectively, these execution outputs provide empirical evidence of stable system operation, correct pipeline integration, and reproducible evaluation behaviour.

```
Training RS-3 Fold 2/5
  - LSTM
  - XGBoost
  - LightGBM
  - Random Forest
  Results: Acc=1.0000, F1=1.0000, AUC=1.0000

Training RS-3 Fold 3/5
  - LSTM
  - XGBoost
  - LightGBM
  - Random Forest
  Results: Acc=1.0000, F1=1.0000, AUC=1.0000

Training RS-3 Fold 4/5
  - LSTM
  - XGBoost
  - LightGBM
  - Random Forest
  Results: Acc=0.9535, F1=0.9730, AUC=0.9775

Training RS-3 Fold 5/5
  - LSTM
  - XGBoost
  - LightGBM
  - Random Forest
  Results: Acc=0.9286, F1=0.9610, AUC=0.9459

------------------------------------------------------------
```

Figure 4.11: Training progress output across cross-validation fold

```
Model Comparison Summary
================================================

           Model  Train Accuracy  Test Accuracy  F1 Score    Recall  Precision
0        XGBoost        0.771429       0.516667  0.485479  0.516667   0.488148
1       LightGBM        0.792857       0.500000  0.487662  0.500000   0.485000
2  Random Forest        0.885714       0.650000  0.643369  0.650000   0.646032


Testing on Custom Sample Data - Including Challenging Cases
================================================

Sample Test Data:
   Temperature (°C)  Systolic_BP (mmHg)  ...  Battery_Level (%)  Sensor_Type_Encoded
0              37.2                 120  ...                 85                    1
1              38.9                 145  ...                 45                    2
2              36.1                 110  ...                 92                    0
3              37.5                 135  ...                 60                    1
4              39.2                 150  ...                 30                    2
5              36.8                 115  ...                 88                    0
6              38.0                 130  ...                 55                    1
7              37.0                 125  ...                 78                    2
8              37.8                 138  ...                 50                    1
9              38.5                 142  ...                 40                    2

[10 rows x 9 columns]
```

Figure 4.12: Model Comparison and Sample Testing

```
 Analyzing patient vitals...
   Temperature trending up: 37.2°C → 38.5°C
   Blood pressure increasing: 120/80 → 155/95
   Heart rate accelerating: 75 → 105 bpm
   Risk factors: Hypertension stage 1, Tachycardia, Fever

  Models are trained and saved!
  Model location: C:\Users\surya\OneDrive\Desktop\thesis

======================================================================
NEXT STEPS: CREATE PREDICTION SCRIPT
======================================================================

======================================================================
 TRAINING COMPLETE!
======================================================================
Training Accuracy: 97.5%
Ready for deployment and predictions
```

Figure 4.13: Patient Vital Analysis Output

4.7 Error Analysis

To validate the proof-of-concept framework, systematic error analysis was conducted on misclassified samples. False negatives (at-risk cases classified as normal) constituted approximately 0.4% of predictions and predominantly occurred during early-stage deterioration with subtle vital sign deviations. False positives (normal cases classified as at-risk) represented approximately 17.3% in certain folds, typically resulting from transient spikes and the conservative ensemble design prioritizing sensitivity. Analysis revealed that cases with ambiguous physiological patterns and boundary-region vital signs exhibited higher error rates. These findings inform

future directions including personalized baseline adaptation, enhanced preprocessing for real-world sensor noise, extended temporal windows, confidence scoring, and multi-stage risk assessment. The analysis demonstrates strong proof-of-concept performance while identifying limitations that provide foundation for future clinical development with real-world validation.

4.8 Chapter Summary

In this chapter, an extensive experimental assessment of the smart health risk-predictive articulations provided by the smart healthAI framework was presented for mobile Internet of Medical Things (IoMT)-based vehicular visioning. Static machine learning methods were observed to have shortcomings via a systematic analysis, including dataset exploration, baseline model comparisons, training behaviour analysis, and multiple metrics to assess performance. Our experiment showed that classical models do not adequately capture temporal patterns in health or differences between patients, which is why all the approaches under perform on tests that involve continuous monitoring.

Comparatively, the LSTM-based ensemble framework achieved superior performance across all evaluation metrics, including accuracy, F1-score, sensitivity, and AUC. Temporal modelling, attention mechanisms, and weighted ensemble fusion contributed to consistent performance across evaluation metrics under continuous monitoring conditions. Our evaluation, through comparative analysis with existing IoMT-AI studies, further validated SmartHealthAI as a high-performing, generalizable solution for real-time mHealth monitoring.

In general terms, the results of this chapter firmly support the design decisions underlying the proposed framework and provide a solid basis for the discussions, implications for practice, and future research posited in the subsequent chapter.

Chapter 5

DISCUSSION

5.1 Overview of Discussion

This chapter presents a critical discussion of the experimental findings obtained from the SmartHealthAI framework, with direct reference to the results reported in Chapter 4. Rather than reiterating numerical performance values, the discussion interprets observed performance trends to explain why the proposed framework performs effectively and how it addresses key limitations of existing IoMT-based health monitoring systems.

The current study concludes health-risk prediction as a binary classification task (Normal vs At-Risk) as a purposeful proof-of-concept design choice. The objective is to validate the feasibility of real-time IOMT risk prediction under controlled experimental conditions rather than providing comprehensive clinically diagnostic . Extensions toward multi-level risk scoring and personalised health prediction are considered future work once the framework is validated using real-world sensor data.

The use of binary classification in this study is a deliberate proof-of-concept design choice. This decision is motivated by three factors. First, the lack of real patient data with clinically validated severity labels makes multi-class classification unreliable in a simulated environment. Second, binary classification provides a clear and controlled framework for validating the core components of the model, including temporal modelling and ensemble learning. Third, in remote patient monitoring, the primary objective is the timely detection of any abnormal physiological condition. While multi-class risk stratification would improve clinical applicability, it requires access to real, clinically annotated data. Therefore, extending the model to multi-level risk prediction is identified as future extension of the SmartHealthAI framework, as outlined in Section 6.5.

The discussion is explicitly structured around the research questions (RQ1–RQ3) defined in Chapter 1. Each section integrates thematic analysis including baseline model limitations, temporal modelling, ensemble learning, interpretability, early-warning capability, and system-level feasibility within the context of the corresponding research question. This structure ensures a clear and traceable linkage between experimental observations, methodological choices, and research objectives.

5.2 Discussion of RQ1: Feasibility of Simulated IoMT Data for Real-Time Health Risk Prediction

RQ1: *what extent can simulated IoMT physiological data support accurate real-time health-risk classification within a controlled monitoring framework?*

The experimental findings presented in Chapter 4 provide strong evidence that simulated IoMT data can support real-time health risk prediction. The controlled simulation environment enabled the generation of continuous, time-stamped physiological streams that exhibit realistic variability, inter-signal correlation, and gradual deterioration patterns consistent with longitudinal health monitoring.As demonstrated in table4.2, the framework achieved an accuracy of 97.5% (±2.3%), F1-score of 98.6% (±1.3%), ROC-AUC of 93.8% (±6.7%), and most critically, a sensitivity of 99.6%, confirming that simulation-based physiological data can effectively substitute for physical sensor data during development and validation phases.

Results from Chapter 4 demonstrate that SmartHealthAI successfully identified progressive health risk trends without reliance on proprietary wearable hardware. This confirms that accurate health risk prediction does not necessarily depend on physical sensors during the development and validation stages. Instead, simulation-driven IoMT environments provide a reproducible, scalable, and cost-effective platform for algorithm evaluation and system benchmarking. The exploratory data analysis in Section 4.2 confirmed data realism, with Figure 4.1 showing gradual variations in systolic blood pressure (120-168 mmHg) and heart rate (78-118 bpm) over 24 hours. Figure 4.2 and Figure 4.3 revealed overlapping distributions between Normal and At-Risk states, reflecting realistic clinical ambiguity. Feature importance analysis in Figure 4.8 and table4.5 showed that temporal statistical features dominated the top 20 predictors, validating that LSTM layers successfully captured time-dependent patterns.

Importantly, the framework showed sensitivity to progressive deviations in physiological parameters rather than reacting only to extreme threshold violations. Sample-level analyses revealed that risk escalation occurred gradually in response to sustained abnormal trends, indicating that simulated data when structured temporally preserves clinically meaningful information necessary for early detection. Sample-level analyses in table4.3 and the practical testing scenario in Section 4.4 demonstrated this capability: the system correctly classified a gradually deteriorating patient as High Risk with 84.2% confidence (temperature rising 37.2°C→39.1°C, systolic BP 125→168 mmHg, heart rate 78→118 bpm over five time points) before reaching extreme thresholds.

Discussion Outcome (RQ1):

The quantitative evidence from table4.2-4.4. achieving 97.5% accuracy, 99.6% sensitivity, and 32.5 percentage point improvement over baseline models that achieved only 51.7-65% accuracy as shown in table4.1, combined with Figures 4.1-4.8 demonstrating stable cross-validation performance in Figure 4.5-4.6 and realistic temporal patterns, combined with scenario-based validation achieving 84.2% Risk Probability, conclusively demonstrates that simulated smartwatch data is a scientifically credi-

ble and practically viable foundation for designing, testing, and validating real-time IoMT-based health monitoring systems, provided that subsequent real-world validation is conducted prior to clinical deployment.

5.3 Discussion of RQ2: Effectiveness of HealthRiskNet Compared to Baseline Models

RQ2: *How effective does the proposed HealthRiskNet framework perform in classifying patient health status compared to baseline ML models in terms of accuracy, sensitivity, and temporal detection capability?*

Baseline models relying on static physiological features demonstrated limited predictive capability, as observed in Chapter 4.3.1. These models frequently misclassified early-stage deterioration as normal and reacted only after substantial deviations occurred. This behaviour highlights a fundamental limitation of snapshot-based learning: the inability to represent temporal continuity in physiological processes. As quantified in table 4.1, baseline models demonstrated limited predictive capability, with XGBoost achieving only 51.7% accuracy, LightGBM 50%, and Random Forest 65% all with F1-scores below 0.64. These results, significantly below the proposed framework's 97.5% accuracy in table 4.2, highlight the fundamental limitation of snapshot based learning in capturing temporal continuity in physiological processes.

Physiological deterioration rarely occurs abruptly; instead, it evolves through subtle, correlated changes across multiple vital signs. Static models treat each observation independently and therefore fail to capture evolving health trajectories. This limitation was evident in the confusion matrices and temporal risk plots presented in Section 4.3.1, where baseline models lacked sensitivity to gradual changes. The comparative analysis in table 4.4 shows a 32.5 percentage point accuracy improvement and demonstrates that the proposed framework provides explicit temporal modeling, patient adaptation capability, and very high sensitivity features entirely absent in baseline approaches.

HealthRiskNet directly addresses in section 3.6 this limitation by integrating temporal modelling through LSTM layers. Sequential learning enabled the model to distinguish normal physiological variability from clinically meaningful deterioration patterns. The attention mechanism further refined this process by assigning greater importance to time segments exhibiting sustained or accelerating deviations, thereby enhancing early-stage sensitivity. This is evidenced by the feature importance analysis in Figure 4.8 and Table 4.5, where temporal features (heart rate variability, blood pressure trends) dominated the top 20 most influential predictors. The ROC curves in Figure 4.5 demonstrate consistent discrimination capability across all cross validation folds with mean AUC of 93.8% ($\pm$6.7%), while confusion matrices in Figure 4.6 show stable classification with minimal false negatives.

The ensemble learning strategy outlined in Section 3.7 and evaluated in Section 4.3.3 achieved an F1-score of 98.6% ($\pm$1.3%) and sensitivity of 99.6%. This reduced false negatives to only 0.4% of predictions an especially critical improvement in healthcare

where missing At Risk cases can delay treatment. The training dynamics shown in Figure 4.7 demonstrate stable convergence without overfitting, confirming that the ensemble approach provides robust generalization across diverse physiological patterns.

Discussion Outcome (RQ2):

HealthRiskNet significantly outperforms baseline models, as evidenced by the 32.5 percentage point accuracy improvement from 65% maximum baseline to 97.5%, 98.6% F1-score, 99.6% sensitivity as shown in Tables 4.1-4.4, and consistent cross-validation performance in Figures 4.5-4.7. This superiority stems from its alignment with the temporal nature of physiological processes and integration of ensemble decision fusion to improve reliability, robustness, and generalization.

5.4 Discussion of RQ3: Mobile Feasibility, Interpretability, and Scalability

RQ3: *:How does the proposed simulated IoMT–AI framework perform with respect to real-time feasibility, interpretability, and scalability in continuous health-monitoring scenarios?*

The smartphone-centric design of SmartHealthAI proved effective for real-time health monitoring. By consolidating data acquisition, preprocessing, inference, and visualisation within a single mobile workflow, the framework avoids latency, connectivity dependence, and architectural complexity associated with cloud or fog-centric solutions. System-level evaluations reported in Section 4.6, with execution evidence shown in Figures 4.11-4.13, confirmed stable execution and low-latency inference on mobile-compatible platforms running Windows 11 with standard hardware (8GB RAM, Intel Core i5 processor) as detailed in Section 5.8. The training completion across all cross-validation folds in Figure 4.11 and successful real-time patient vital analysis output from Figure 4.13 demonstrate that advanced temporal deep learning models can be deployed efficiently within resource-constrained mobile environments when designed appropriately.

Interpretability is a key strength of the proposed framework. Attention-based temporal reasoning and feature-level analysis enabled predictions to be linked to meaningful physiological trends rather than opaque model decisions. As shown in Figure 4.8 and Table 4.5, the top 20 most influential features are clinically interpretable temporal statistics with clear physiological significance. The practical testing output in Section 4.4 and Figure 4.13 demonstrates that the system provides both risk probability (84.2% in the High Risk scenario) and confidence scores, supporting transparent decision-making. Unlike post hoc explainability methods, interpretability in SmartHealthAI is embedded directly into the learning process through the attention mechanism, supporting transparency, accountability, and clinician trust.

From a scalability perspective, the simulation-driven approach enabled extensive experimentation across stratified 5-fold cross-validation and achieving mean accuracy 97.5% ±2.3% with low standard deviation, as shown in table 4.2, demonstrating

stability across diverse physiological conditions without additional hardware costs. The fold-wise stability shown in confusion matrices in Figure 4.6 and consistent learning curves in Figure 4.7 confirm that the framework generalizes reliably, making SmartHealthAI particularly suitable for large-scale deployments and low-resource healthcare settings.

Discussion Outcome (RQ3):

SmartHealthAI effectively balances real-time performance 97.5% accuracy, inference on standard mobile hardware as evidenced in Figures 4.11-4.13, interpretability and attention based feature importance shown in Figure 4.8 and table 4.5, with 84.2% confidence scoring in practical scenarios, and scalability, stable 5-fold cross-validation performance with ±2.3% standard deviation across table 4.2, Figure 4.6, demonstrating strong readiness for practical mobile IoMT deployment.

5.5 Clinical Relevance and Early-Warning Capability

A major implication of the experimental findings in chapter4 demonstrates behaviour consistent with an early-warning system. Case-based validation in Chapter 4 showed that HealthRiskNet classified At-Risk conditions based on cumulative temporal evidence rather than isolated abnormal readings.

This behaviour closely mirrors clinical reasoning, where trends and progression carry greater diagnostic significance than single measurements. By modelling health deterioration as a dynamic temporal process, SmartHealthAI aligns computational inference with real clinical workflows.

The probabilistic risk outputs further enhance clinical usability. Continuous risk scores allow prioritisation of alerts, reduce alarm fatigue, and support informed decision-making rather than binary alarm generation.

5.6 Comparison with other IoMT-AI studies

table 5.1 provides a comparative summary of representative IoMT–AI studies and the proposed SmartHealthAI framework with respect to application focus, modelling strategies, temporal learning capability, evaluation scope, and system-level limitations. As indicated in the comparison, a significant amount of current research is focused on disease- or application-specific use cases, including mobile stroke prediction, obesity, cardiovascular disease, diabetes, and neurodegenerative disease monitoring. Though these methods show strong performance in their domain, they are limited by their specialised design and cannot be widely applied to monitor other health conditions. Their extensibility is also low across different physiological variables.

Most of the studies reported in Table 5.1 primarily rely on static feature analysis or utilise only partial temporal modelling. Many of these frameworks cannot truly learn temporal health trends as they treat observations as independent samples.

Thus, these systems are not capable of detecting subtle declines in health but are better at detecting abrupt health transitions. For example, in a mHealth setting, patients' health status changes continuously throughout the day.

The capabilities of recurrent and hybrid neural architectures for health analytics are well-received in recent deep learning–based IoMT systems. Nonetheless, as shown in Table 5.1, most of these frameworks assess single models separately and do not adequately address robustness across different physiological patterns or operational conditions. Such a limitation is more pronounced in non-ensemble-based approaches due to higher sensitivity to data noise, superficial variations, and data variability.

SmartHealthAI overcomes these limitations by employing an ensemble learning strategy that combines temporal deep learning with multiple complementary machine learning classifiers. The framework incorporates various predictive viewpoints and ensures the stability and reliability of health risk prediction. While ensemble-based techniques are explored in the context of IoMT, their use for patient-centric health risk prediction is still relatively rare in the existing literature.

At a system level, as shown in Table 5.1, some existing frameworks rely on purely cloud-centric, edge-based, or fog-based architectures. Although these designs enhance scalability and governance, they add architectural complexity and can make deployment to resource-constrained mobile platforms more difficult. Instead, SmartHealthAI employs a smartphone-centered architecture that integrates data acquisition, processing, and analysis into a single workflow.

A third differentiator revealed in this comparison is interpretability. Although some IoMT studies propose explainable AI techniques, they typically use them as auxiliary tools rather than tightly integrating them into the analytical process. The SmartHealthAI framework incorporates attention-based temporal learning with feature-level reasoning to link health risk predictions to clinically interpretable physiological trends.

Lastly, supported by Table 5.1, the comparative analysis suggests that SmartHealthAI provides a coherent, temporally aware, and resilient health-monitoring solution. The proposed framework incorporates temporal modelling, ensemble learning, interpretability, and mobile-centric system design, addressing key research gaps reported in previous literature.

5.7 Validation Strategy and Threats to Validity

5.7.1 Validation Strategy

Each stage of system design, modelling, and evaluation employed several validation strategies to ensure reliability, robustness, and credibility.

Controlled simulation-based validation was employed to remove confounding variability arising from proprietary wearables. In a simulated smartwatch environment, physiological data streams were produced under well-defined and repeatable conditions, allowing systematic experimentation and replication.

Table 5.1: Comparison of IoMT-based health risk prediction studies

Study	**Application Domain**	**Modeling Approach**	**Temporal Modeling**	**Key Metrics Reported**	**Best Reported Performance**	**Limitations**
Elbagoury et al. [36]	Mobile stroke prediction	CNN + GMDH-LSTM	Yes	Accuracy, AUC	Accuracy = 94.2%	Single disease focus, no ensemble
Jeong et al. (DeepHealthNet) [50]	Adolescent obesity prediction	Deep Neural Network	Limited	Accuracy, F1	Accuracy = 91.8%	Static-dominant features
Islam et al. [49]	Cardiovascular risk prediction	ML + IoT sensors	No	Accuracy	Accuracy = 88.6%	No temporal learning
Rodríguez-Rodríguez et al. [58]	Diabetes management	Wearable-based ML	Partial	AUC, Precision	AUC = 90.1%	Disease-specific, limited generalisation
Goumidi & Pierre [30]	IoMT anomaly detection	Stacking ML models	No	Accuracy, Recall	Accuracy = 92.4%	Focused on network anomalies
Sandulescu et al. [60]	Neurodegenerative monitoring	ML-based prediction	Partial	Accuracy	Accuracy = 89.7%	Limited real-time evaluation
Yadav et al. [32]	Ambient assisted living	DNN-based IoMT	No	Accuracy	Accuracy = 90.3%	No patient-level temporal modelling
Proposed SmartHealthAI	**Mobile IoMT health risk prediction**	**LSTM + Attention + Ensemble**	**Yes**	**Accuracy, F1, AUC, Sensitivity, Specificity**	**Accuracy = 97.5%, F1 = 98.6%**	**Validated; real-time extension possible**

Temporal validation was achieved by modelling biological pathways using sequential time windows rather than independent observations. This design enables modelling gradual patterns of health deterioration.

Stratified k-fold cross-validation was performed to provide statistical robustness and generalisation across different data partitions, reducing bias and overfitting.

The proposed HealthRiskNet model was comparatively validated against baseline approaches, providing quantitative validation of hybrid temporal–statistical modelling and ensemble fusion.

End-to-end system validation evaluated real-time feasibility, latency, and execution robustness on smartphone-admissible platforms, ensuring deployability in mHealth monitoring scenarios.

Combined, these validation approaches position SmartHealthAI as a reproducible and practical system for real-time health risk prediction in mobile IoMT scenarios.

5.7.2 Threats to Validity

Although a rigorous approach was adopted, several threats to validity must be considered.

• Construct Validity:

The framework utilises simulated physiological data. While this supports controlled experimentation and reproducibility, synthetic data may not fully represent real-world sensor noise, artefacts, or behavioural variability.

• Internal Validity: The health risk prediction task is modelled as a binary clas-

sification problem (Normal vs. At-Risk). This abstraction supports early warning systems but does not capture disease-specific severity levels.

• External Validity: Evaluations were conducted on simulated and public datasets that may lack population diversity, long-term behavioural change, and device heterogeneity observed in real-world deployments.

• Ecological Validity: Experiments were conducted without clinician-in-the-loop or patient-in-the-loop feedback, meaning real clinical usage dynamics are not fully represented.

• Technological Validity: Although smartphone feasibility was assessed, variations in hardware, operating systems, and background processes may affect real-world latency and performance.

While these threats do not invalidate the framework, they provide guidance for interpreting results and for future clinical validation. Present findings support regarding feasibility of real-time interface, effectiveness and performance under controlled experimental conditions. For real-world clinical deployment and resilience to sensor noise should be validated with real wearable data which clearly clarifies the scope of our current evaluation and future direction .

5.8 System Requirements for SmartHealthAI

To clarify smartphone deployment assumptions and operational feasibility, the system requirements and execution specifications are summarized below. All experiments and evaluations were carried out on the Windows 11 (64-bit) platform.

Development and Execution Environment:

- Operating System: Windows 11 (64-bit)
- Processor: Intel Core i5 or equivalent multi-core processor
- RAM: 8 GB system memory
- Storage: Approximately 20 GB free disk space for dataset storage, model execution, and result logging
- Python version: 3.8 or higher
- Key libraries: TensorFlow with Keras, NumPy, Pandas, Scikit-learn

Inference Execution Context:

- Location: Desktop environment during development and validation
- Target deployment: Smartphone-based mobile platforms
- Hardware acceleration: Not required
- Processing capability: Sufficient for physiological signal preprocessing, HealthRiskNet inference, and real-time health risk classification

Deployment Assumptions:

- Framework designed for smartphone-centric execution
- Mobile deployment feasibility demonstrated through resource-constrained testing
- No dedicated GPU required, supporting applicability to resource-bounded mobile IoMT healthcare settings
- Implementation uses Anaconda scientific computing environment for robustness and reproducibility

In summary, the SmartHealthAI framework operates on standard computing hardware without specialized acceleration requirements. The minimal resource demands (8GB RAM, standard multi-core CPU) demonstrate feasibility for deployment on modern smartphones and edge platforms, supporting the mobile-centric healthcare monitoring objectives of this research.

5.9 Summary of Discussion

This chapter provided a critical discussion of the experimental results of the SmartHealthAI framework in relation to the research questions and outcomes presented in Chapter 4. The discussion demonstrated that simulated IoMT data can facilitate accurate real-time health monitoring, that temporal and ensemble learning outperform static baseline techniques, and that a smartphone-based architecture can deliver interpretable and scalable health risk prediction.

The overall design and comprehensive validation effort position SmartHealthAI as an important step forward for IoMT-enabled mHealth monitoring. The insights from this discussion lay the groundwork for the conclusions, limitations, and future research directions presented in the next chapter.

Chapter 6

Conclusions and Future Work

6.1 Introduction

This chapter provides an overall conclusion of the research presented in this thesis by summarising the key findings, assessing the aims achieved, and reflecting on the broader contribution of the SmartHealthAI framework to Internet of Medical Things (IoMT)- enabled healthcare systems. The main driving force behind this work was the need to overcome the key shortcomings of current IoMT-AI solutions, namely, their non-temporal nature, poor resilience to dynamic physiological conditions, lack of interpretability, and practical deployment on mobile platforms.

The methodological, experimental, and analytical foundations of the thesis were established in chapters 3, 4, and 5. The architectural design of the SmartHealthAI framework, its data acquisition pipeline, preprocessing strategy, feature engineering process, and the HealthRiskNet model are described in Chapter 3. The rigorous experimental evaluation in Chapter 4 has shown consistent learning behaviour, effective temporal modelling, and stable classification outcomes across different evaluation settings. These findings were contextualised in Chapter 5 through an in-depth discussion and comparison with existing IoMT-AI studies, showcasing the benefits of temporal learning, ensemble decision fusion, and smartphone-centric system design.

6.2 Summary of the Work and Key Findings

This thesis proposes an intelligent Internet of Medical Things (IoMT) framework, SmartHealthAI, to enable continuous, real-time health risk prediction through artificial intelligence (AI) on mobile platforms. This was a framework devised as a response to the increasing need for advanced healthcare solutions that can track physiological signals over time and provide behavioural drugs when there are signs that a risk of an adverse event may occur, which may lead to severe disease.

The core of SmartHealthAI is the HealthRiskNet model, a temporal deep learning model with attention mechanisms for capturing the dynamics of physiology. Wherein classical machine learning approaches model each observation independently, HealthRiskNet models the sequential dependencies of physiological data, enabling this system to identify trends, anomalies, and physiological drift from normal

health states. In healthcare, where deterioration rarely arises from isolated abnormal readings but instead progresses, this temporal perspective is essential.

Beyond the temporal learning component, the framework integrates static and contextual physiological features to retain records of static health status. By combining temporal and static representations, we account for both short-term physiological changes and long-term trends during prediction. To improve reliability, we use an ensemble learning strategy that combines forecasts from both approaches. This ensemble method reduces the impact of noise and the influence of model-specific biases, thereby increasing robustness across various physiological states.

Meanwhile, The proposed framework displayed stable convergence across all cross-validation folds with training and validation loss curves consistently aligned as shown in Figure 4.7, achieving 97.5% accuracy (±2.3%) and 99.6% sensitivity as shown in Table 4.2. The ROC curves in Figure 4.5 demonstrated predictable discriminative behavior with mean AUC of 93.8% (±6.7%) across folds, while fold-wise confusion matrices as shown in Figure 4.6 confirmed greater sensitivity to health risk situations with false negative rates of only 0.4% compared to baseline models' inability to reliably detect At-Risk cases by achieving only 51.7-65% accuracy in Table 4.1. This emphasised interpretability as a design principle, not merely a post-hoc feature. SmartHealthAI integrates attention-based temporal reasoning with feature-level explanation mechanisms shown in Figure 4.8 and Table 4.5, where the top 20 most influential features are clinically interpretable temporal statistics. The practical testing scenario in Section 4.4 demonstrates this transparency: the system provided both risk classification to at Risk and confidence scoring 84.2% probability with clear physiological rationale as the temperature rising from 37.2°C to 39.1°C, BP changes from 125 to 168 mmHg, HR 78→118 bpm, enabling clinical validation of the decision pathway. This transparency improves trust and usability in clinical or decision-support situations. The quantitative evidence—97.5% accuracy, 98.6% F1-score, 32.5 percentage point improvement over baselines as shown in Table 4.4, stable cross-validation performance in Figures 4.5 and 4.7, and interpretable feature attribution in Figure 4.8 confirms the efficacy of the proposed methodology and demonstrates its viability for mobile IoMT-based health monitoring.

6.3 Research Contributions and Practical Significance

This thesis, therefore, contributes to a deeper understanding of IoMT-enabled intelligent healthcare in several substantial ways. The first contribution is an overview of the unified SmartHealthAI framework that combines aspects of data acquisition and preprocessing, such as temporal modelling, ensemble learning, and interpretable decision-making, within a single smartphone-centric architecture.

The HealthRiskNet architecture illustrates how temporal modelling using LSTM and attention mechanisms can effectively model complex physiological dynamics. Such limitations of conventional IoT models that rely on static or short-term analysis, thereby neglecting long-range dependencies in health data, can ultimately be addressed by this approach.

Finally, the ensemble learning component is another significant contribution. The framework increases robustness and generalisation by fusing predictions from multiple models, which is beneficial for safety-critical healthcare applications.

At the level of practical significance, SmartHealthAI shows excellent promise as a mHealth decision-support tool. The smartphone-centered design enables implementation in real-world settings, including low-resource environments where access to sophisticated medical infrastructure might be limited. Result: The framework can help healthcare providers with continuous monitoring of patient health status and the provision of actionable, interpretable risk forecasts that can trigger early intervention.

In addition, the focus on interpretability brings the framework in line with ethical and regulatory requirements for medical AI systems. SmartHealthAI enables clinical validation with precise, understandable predictions that facilitate error analysis among clinicians and enhance operator trust. The clear scoping between the classification-focused research prototype and future system extensions enhances the rigor with which our contributions are evaluated, whilst providing a solid basis for these extensions.

6.4 Conclusions

Predicting health risk in mobile IoMT environments is a complex problem, as models must be temporally aware, robust, interpretable, and deployable. The present thesis tackled these challenges by proposing and evaluating the SmartHealthAI framework, which incorporates temporal deep learning, ensemble decision fusion, and explainable analytics into a single-system mobile healthcare approach.

Here, we show the explicit need for temporal modelling for sound health risk prediction. SmartHealthAI uses sequential learning and attention mechanisms to identify important temporal patterns and prioritise clinically relevant information.

Then, interpretability becomes one of the cornerstones of trusted medical AI. SmartHealthAI embeds explainability within the modelling process, avoiding black-box decision-making and providing a clear understanding of model behaviour. Transparency in the system supports clinician trust and collaborative decision-making and ensures adherence to healthcare ethics.

This thesis thus shows that combining temporal intelligence, ensemble robustness, and a mobile-system design can lead to a better-performing and more reliable system for health monitoring using IoMT. The SmartHealthAI framework is a significant contribution to an intelligent mHealth system, suited to address access and accountability issues and to provide multiple insights for continued research in IoMT and AI-driven health analytics.

6.5 Future Scope

While results demonstrate strong performance in controlled simulation environments, clinical validation with real patient data from actual wearable sensors is essential before practical deployment. Several promising directions remain for future research and enhancement:

- **Multi-Class and Condition-Specific Health Risk Prediction**

  Future work can extend the current binary health-risk classification framework to multi-class or disease-specific risk assessment models. Such an extension would enable more granular clinical interpretation and broader applicability across chronic disease management and long-term health surveillance scenarios.

- **Integration of Multimodal Physiological and Contextual Data**

  The framework can be enhanced by incorporating additional physiological signals and contextual information, including physical activity, sleep patterns, and environmental factors. Multimodal data fusion has the potential to improve predictive robustness and clinical relevance. Furthermore, leveraging individual patient baselines alongside adaptive and personalised modelling strategies can improve sensitivity while reducing false alerts.

- **Privacy-Preserving and Decentralised Learning Mechanisms**

  Future research may integrate federated learning techniques to support privacy-preserving model training across distributed devices while keeping patient data local. Additionally, blockchain-based mechanisms may be explored to strengthen data integrity, access control, and trust management within distributed IoMT environments.

- **Lightweight and Energy-Efficient Mobile Deployment**

  At the system level, future work should investigate model optimisation techniques such as lightweight architectures, model pruning, adaptive sampling, and edge-optimised inference strategies. These enhancements would support long-term deployment on resource-constrained mobile devices while improving scalability and energy efficiency.

- **Clinical Validation and Real-World Deployment**

  A critical future direction involves real-world validation through clinician-in-the-loop evaluations and pilot deployments using live sensor data. Assessing usability, reliability, and clinical acceptance, as well as aligning explainability outputs with clinical reasoning, will be essential for safe and effective integration into healthcare workflows.

- **Comprehensive Validation, Ethical Framework, and Regulatory Pathway**

Beyond the technical enhancements outlined above, future work must address comprehensive error analysis with real-world data to characterise misclassification patterns and failure modes, along with ethical framework development encompassing privacy protection, data governance, and informed consent. Implementation should follow a phased roadmap: short-term objectives (6-12 months) including real sensor pilot testing and error tracking; medium-term goals (1-2 years) addressing multi-class risk stratification, clinical validation studies, and EHR interoperability; and a long-term vision (3-5 years) pursuing randomised controlled trials, regulatory approval, and integration into telemedicine platforms. Achieving these goals will require interdisciplinary collaboration, diverse clinical datasets, sustained funding, and institutional partnerships.

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Faculty of Computing, Blekinge Institute of Technology, 371 79 Karlskrona, Sweden